



Food and Agriculture
Organization of the
United Nations

Global Forest Resources Assessment 2020

Report

China

Rome, 2020



FAO has been monitoring the world's forests at 5 to 10 year intervals since 1946. The Global Forest Resources Assessments (FRA) are now produced every five years in an attempt to provide a consistent approach to describing the world's forests and how they are changing. The FRA is a country-driven process and the assessments are based on reports prepared by officially nominated National Correspondents. If a report is not available, the FRA Secretariat prepares a desk study using earlier reports, existing information and/or remote sensing based analysis.

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Introduction

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Introductory text

The Global Forest Resources Assessment (FRA) is a global periodical forest resources assessment activity led by the FAO and participated in by its member countries. The results are important basis to inspect the intergovernmental process of sustainable forest management and assess the compliance of all member countries with international conventions. At the 1st FAO Annual Meeting held in 1945, consensus was reached on “acquiring the latest information of global forest resources”. According to relevant provisions of the “Constitution of the Food and Agriculture Organization of the United Nations”, FAO organized the first Global Forest Resources Assessment (FRA) in 1947. At the 6th FAO Annual Meeting held in 1951, a request for regular and continuous obtaining information of global forest resources every five or ten years was proposed. Currently, FAO has organized 13 FRAs and the ongoing FRA2020 is the 14th assessment.

The Chinese government has always attached great importance to the Global Forest Resources Assessment. Since 2000, according to FAO's requirements, the National Correspondent has been designated by the former State Forestry Administration to participate in four FRAs at the government level, namely FRA2000, FRA2005, FRA2010 and FRA2015, and country reports that reflect the latest status of forest resources and forestry development in China have been prepared and submitted to FAO. The forest resources assessment work conducted by China has been highly recognized by FAO.

The National Forestry and Grassland Administration attaches great importance to the FRA2020, and relevant departments and bureaus are designated to undertake specific assessment tasks according to their business functions. According to FAO's schedule for the FRA2020 as well as actual practice of forest resources survey & monitoring and forestry statistical work in China, since April 2018, information collection, data analysis and assessment, preparation and submission of draft country report, review and revision of draft country report have been accomplished. Now the final country report has been completed, provided the basis for FAO's assessment of forest resources, forest protection activities and forestry development achievements in China.

1. Status of Forest Inventory & Monitoring and Forestry Statistics in China

Since the 1970s, China has conducted national forest inventory every 5 years. Adopting the internationally recognized method of “continuous forest inventory (CFI)” and using provincial (autonomous regions, municipalities directly under the Central Government, hereinafter referred to as province) surveys as a whole, a systematic sampling method was used to establish and verify fixed sample plots (a total of 415,000 ground sample plots in the country). In addition, modern techniques such as remote sensing have been adopted to obtain the status of forest resources of the country and provinces as well as their dynamic changes in a five-year period, and to assess the status of forest resources and their functional benefits. Up to now, 8 national forest inventories (NFIs) have been completed. Previous NFI results provide a scientific basis to accurately grasp China's forestry development trends, formulate and adjust forestry guidelines and policies, and compile key national and economic development plans. The NFI results also become an important window for displaying China's achievements in forestry development and ecological improvement. The results also meet the information needs of forestry development and economic development in different periods, and play an important role in China development.

In addition, according to the needs of China's forestry development, the nationwide monitoring of desertification land, wetland resources and forest fires, as well as survey on forest pests, and specific investigation on forest resources management has been carried out, resulting in a relatively complete surveying technology system and institutional management system. At the same time, forestry statistical work is carried out annually and the "National Forestry Statistical Yearbook" is published every year. Relevant departments also combined their own practices and regularly collected data from the lower forestry authorities to grasp the overall status of the country.

The national forest inventory, related forestry monitoring and statistical works provide detailed and basic data for national reporting to the FRA2020 as well as compiling the country reports. Forest resources inventory and monitoring as well as forestry statistical works relevant to the FRA2020 are shown in Table 1.

Table 1 Forest inventory & monitoring and forestry statistics in China

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Monitoring schemes	Objectives	Methods	Indicators	Implementation
National Forest Inventory	To periodically re-survey the status quo of forest resources at national and provincial levels, and identify the macro-change in terms of growth and consumption of forest resources.	Field measurement of ground sampling plots based on sampling survey techniques.	About 160 indicators about the quantity, quality, structure and functions of forest resources.	9 NFIs finished
Forest fire monitoring	To monitor the happening and expanding trend, warn and forecast forest fire.	Combination of satellites remote sensing, aircraft patrolling, watchtower and ground patrolling.	Fire type, happening region, burned area and fire rank, etc.	Continuous real time monitoring. Information collection by month and year.
Monitoring forest biotic agents	To monitor and early forecast the happening and expanding of forest insect, pest and rat disasters.	Field observation, inspection and quarantine of wildlife.	Happening, prevention and cure of main types, baleful life-form broadcasting, diffusing and spreading, etc.	Real-time monitoring and annual national report.
Forestry statistics	Identification of the general situation of nationwide forestry activities to figure out the achievement of forestry ecological improvement and industry development.	National information collection from local level to national level.	Forestry industry development, afforestation, key forestry programs, forest disaster, industrial production, education, employment and financial investment, etc.	Annual statistics and publishing.
Department statistics	To identify the general status of work in relation with forestry.	Collection of local reported figures by related department of NFGA.	Resources management, natural reserve establishment, afforestation, prevention and cure of insects and pests, etc.	Periodical or non-periodical statistics.

2. Reporting Process of China for FRA2020

1) Preparatory stage. After the FRA 2020 Launch and Technical Meeting of National Correspondents and CFRQ Partners held in Mexico in March 2018, a FRA2020 working team was established with engagement of 9 departments and institutes (see Appendix). The Chinese assessment team was approved by the NFGA and led by Department of Forest Resource Management and Department of International Cooperation, considering the heavy tasks, vast content, extended fields and involved many functional departments. The "FRA2020: Guidelines and Specifications" and "FRA2020: Terms and Definitions" (hereinafter referred to as "Guidelines and Specifications" and "Terms and Definitions") have been translated, and compiled into technical documents which have been shared with relevant authorities. In accordance with the contents of 7 titles and 22 tables defined in the "Guidelines and Specifications", the departments/bureaus and institutes being responsible for each table are identified, and the assessment tasks and timeline are determined.

2) Draft report stage. Since April 2018, the members of the working team have completed original data collection, data processing, analysis and evaluation, and conducted data estimation and forecasting. A full set of reporting tables and the FRA2020 Country Report of China had been drafted at the middle of June, which was submitted for review to the FRA 2020 Sub-regional workshop for East and South-East Asia held in Tokyo, Japan, 25-29 June 2018. After the workshop, the draft report was revised according to the comments of FAO experts.

3) Final report stage. Since January 2019, the members of the working team have updated the related tables in the draft report based on the 9th NFI results, especially corrected the periodical data of biomass and carbon stock, and presented this final report of China for FRA 2020. During the preparation of the final report, I also communicated with the experts of the FRA project team through the online work platform, and improved the relevant content according to their feedback. The final report was approved by the expert review committee on 28 March. Then, it was signed by the head of National Forestry and Grassland Administration for submission..

3. Compilation for the FRA2020 Country Report of China

Based on China's National Forest Inventory and relevant forestry surveys and monitoring results, as well as forestry statistics, comprehensive analysis on relevant indicators of 7 categories was conducted in accordance with the FRA2020 Guidelines and Specifications, and 22 reporting tables were finally completed.

1) Preparation for national reporting tables

Table 1a, 1b, 1c, 1d, 1e, 2a, 2b, 2c, 2d, 3a, 4a, 4b, 5b 5c, 6a and 6b have been completely finished or mostly finished (without considering the options and a few data gaps); while Table 1f, 3b, 5a, 7a, 7b and 7c have been partly completed (lack of some necessary data). The status of completion of reporting tables is listed in Table 2.

Table 2 Completion status of reporting tables

Title	Tables	Completion status	Remarks
1 Forest extent, characteristics and changes	1a Extend of forest and other wooded land	■	
	1b Forest characteristics	■	
	1c Annual forest expansion, deforestation and net change	■	
	1d Annual reforestation	■	
	1e Specific forest categories	■	
	1f Other land with tree cover	▲	Lack of ome necessary data
2 Forest growing stock, biomass and carbon	2a Growing stock	■	
	2b Growing stock composition	■	
	2c Biomass stock	■	
	2d Carbon stock	●	Lack of data of optional items
3 Forest designation and management objective	3a Designated management objective	■	
	3b Forest area within legally established protected areas and forest area with long-term forest management plan	▲	Lack of some necessary data
4 Forest ownership and management rights	4a Forest ownership	●	Lack of data of optional items
	4b Management rights of public forests	■	
5 Forest disturbances	5a Area affected by disturbances	▲	Lack of some necessary data
	5b Area affected by fire	■	
	5c Degraded forest	■	
6 Forest policy and legislation	6a Policies, legislation and national platform for stakeholder participation in forest policy	■	
	6b Area of permanent forest estate	■	
7 Employment, education and NWFP	7a Employment in forestry and logging	▲	Lack of ome necessary data
	7b Graduation of students in forest-related education	▲	Lack of some necessary data
	7c Production and value of non-wood forest products in 2015	▲	Lack of some necessary data

Note: ■--Fully completed ●--Mostly completed ▲--Partly completed

2) Data sources

The reporting table is filled and the country report is compiled based on the previous National Forest Inventory results, the “China Forestry Statistical Yearbook” since 1990, and the monitoring and statistics data from departments and relevant authorities of the National Forestry and Grassland Administration, as well as other collected data. Information used to prepare the country report includes:

(1) The national forest resources statistics from the 3rd to 9th National Forest Inventory (1984-1988, 1989-1993, 1994-1998, 1999-2003, 2004-2008, 2009-2013, 2014-2018).

(2) The "China National Forestry Statistics Yearbook" since 1990, for national data on planted forest, forestry nature reserves and forest park construction, forest disasters, timber and main forest products, forestry employment, and forestry investment.

(3) Monitoring and statistical data such as forest fire monitoring, forest biotic agent investigation.

(4) Development of forest management plan, sustainable forest management policy and legislation, etc.

(5) China Country Report for FRA 2015.

3) Reporting methodology

According to the Guidelines and Specifications of Global Forest Resources Assessment 2020, national reporting includes the following steps:

(1) Identification and selection of data sources. Based on the requirements of each specific reporting table, all potential data sources are identified and data sources are assessed according to content, completeness, quality, and compliance with FRA definition. Particularly for Table 1a “Extent of forest and other wooded land”, the documentation of the national data sources is extremely important to get an idea of the reliability of the data. For each national data point a category should be assigned that describe the methodology for data collection: National Forest Inventory, remote sensing assessment sample plots, full coverage forest /vegetation map, registers / questionnaires and other source types.

(2) Description of data sources, national classes and definitions, and original data. Firstly, the selected data sources, their references and the corresponding data should be documented according to the full reference (author(s), year of publication (if published) and title), reference year(s) of the data, and necessary additional explanations. Secondly, the national definition of each data source should be clearly stated. In addition, original national data that constitute the basis for the estimates in the table should also be provided. Note that only the original data relevant for each specific reporting table and used for the further analysis need to be provided.

(3) Analysis of national data. The analysis of national data comprises two steps that may or may not be necessary to carry out, depending on the nature of the national data. These include the classification and the estimation and forecasting.

Step 1: Reclassification. The reclassification can be necessary in order to make national data correspond to the categories defined for FRA. When the national classes are identical to the FRA categories, the reclassification can be omitted. Reclassification is performed assigning percentages in a “Reclassification Matrix”, in which each national class is assigned a percentage that applies to each FRA category.

The purpose of reclassification is to ensure the harmonization of national classes. The definition and quantification criteria of most global assessment indicators, including the forest area, are quite different from those in China. Therefore, the existing NFI results cannot be directly involved in the analysis and assessment. During the reporting process to FRA2020, many experts participated in the analysis of the evaluation criteria and definition to ensure the data harmonization, and corresponding relationships between FRA categories and China national classes are established for decomposition and integration.

Step 2: Estimation and Forecasting. The estimation and forecasting of values is often necessary in order to report national data for the FRA reference years. The estimation is the process of interpolation between observations, such as obtaining data of specific year between two NFIs; and forecasting is the extrapolation of values beyond observations, such as obtaining data of year 2020. Estimation and forecasting can be made using linear interpolation or extrapolation. However, in cases where trends are not linear, curvilinear relationships or multiple linear relationships (for different segments of the time series) can be applied.

It is important to stress that estimation and forecasting are not only mathematical calculations. It is equally or even more important to assess whether the estimated/forecasted figures reflect reality. Many times there may be particular reasons why data from different years vary, and such variations do not necessary imply that there is a trend that can be used for estimation and forecasting. If time series indicate trends do not reflect the real situation, this must be documented in the country report. In such cases, countries should make an adjustment of the estimated / forecasted data, and clearly document and justify this in the country report.

It should be noted that, there are items in the reporting tables which China lacks corresponding survey statistics and it is difficult to fill the data in accordance with the FAO definition criteria and requirements, those items will be reported as "ID" (Insufficient Data).

1 Forest extent, characteristics and changes

1a Extent of forest and other wooded land

National data

Data sources

1986	References	State Forestry Administration, 1990, China Forest Resources Statistics (1984-1988)
	Methods used	National Forest Inventory
	Additional comments	Data of 3rd National Forest Inventory in China
1991	References	State Forestry Administration, 1995, China Forest Resources Statistics (1989-1993)
	Methods used	National Forest Inventory
	Additional comments	Data of 4th National Forest Inventory in China
1996	References	State Forestry Administration, 2000, China Forest Resources Statistics (1994-1998)
	Methods used	National Forest Inventory
	Additional comments	Data of 5th National Forest Inventory in China
2001	References	State Forestry Administration, 2005, China Forest Resources Statistics (1999-2003)
	Methods used	National Forest Inventory
	Additional comments	Data of 6th National Forest Inventory in China
2006	References	State Forestry Administration, 2010, China Forest Resources Statistics (2004-2008)
	Methods used	National Forest Inventory
	Additional comments	Data of 7th National Forest Inventory in China
2011	References	State Forestry Administration, 2014, China Forest Resources Statistics (2009-2013)
	Methods used	National Forest Inventory

	Additional comments	Data of 8th National Forest Inventory in China
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2016	References	National Forestry and Grassland Administration, 2019, China Forest Resources Statistics (2014-2018)
	Methods used	National Forest Inventory
	Additional comments	Data of 9th National Forest Inventory in China

Classifications and definitions

1986	National class	Definition
	Arbor forest	Forest land composed of arbor species, with a canopy cover (or crown closure) of more than 20%.
	Economic forest	Forest land composed of economic species, of which main purpose is to produce fruits, cooking oil, drinks, spices, industrial raw materials and medicinal materials.
	Bamboo forest	Forest land composed of bamboo species with a diameter more than 2cm.
	Open forest land	Land composed of arbor species with a canopy cover between 10%-19%.
	Shrub land	Land composed of shrubs and/or bushes with a canopy cover more than 30%, of which main purpose is to cultivate shrubs or used for protection.
	Unestablished afforestation land	Afforestation land that has not been closed but is promising for forest during 3-5 years.
	Nursery land	Land permanently used for cultivating tree seedlings and saplings.
	Cut-over & fired-over forest land	Forest land with a canopy cover less than 10% and within 3 years after clear cut or fired.
	Forest suitable land	Land planned for forestry development by county-level or higher level government.
	Non-forestry land	All land not classified into forestry land above, includes cultivated land, grassland, water area, unused land and construction land.

1991	National class	Definition
	Arbor forest	Forest land composed of arbor species, with forest vegetation and a canopy cover (or crown closure) of more than 20%.
	Economic forest	

		Forest land composed of economic species, of which main purpose is to produce fruits, cooking oil, drinks, spices, industrial raw materials and medicinal materials.
	Bamboo forest	Forest land composed of bamboo species with a diameter more than 2cm.
	Open forest land	Land composed of arbor species with a canopy cover between 10%-19%.
	Shrub land	Land composed of shrubs and/or bushes with a canopy cover more than 30%, of which main purpose is to cultivate shrubs or used for protection.
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	Forest suitable land	Land planned for forestry development by county-level or higher level government.
	Non-forestry land	All land not classified into forestry land above, includes cultivated land, grassland, water area, unused land and construction land.

1996	National class	Definition
	Arbor forest	Forest land composed of arbor species, with forest vegetation and a canopy cover (or crown closure) of more than 20%.
	Economic forest	Forest land composed of economic species, of which main purpose is to produce fruits, cooking oil, drinks, spices, industrial raw materials and medicinal materials.
	Bamboo forest	Forest land composed of bamboo species with a diameter more than 2cm.
	Open forest land	Land composed of arbor species with a canopy cover between 10%-19%.
	Shrub land	Land composed of shrubs and/or bushes with a canopy cover more than 30%, of which main purpose is to cultivate shrubs or used for protection.
	Unestablished afforestation land	Afforestation land that has not been closed but is promising for forest during 3-5 years.
	Nursery land	Land permanently used for cultivating tree seedlings and saplings.
	Cut-over & fired-over forest land	

		Forest land with a canopy cover less than 10% and within 3 years after clear cut or fired.
	Forest suitable land	Land planned for forestry development by county-level or higher level government.
	Non-forestry land	All land not classified into forestry land above, includes cultivated land, grassland, water area, unused land and construction land.

2001	National class	Definition
	Arbor forest	Forest land composed of arbor species, with forest vegetation and a canopy cover (or crown closure) of more than 20%.
	Economic forest	Forest land composed of economic species, of which main purpose is to produce fruits, cooking oil, drinks, spices, industrial raw materials and medicinal materials.
	Bamboo forest	Forest land composed of bamboo species with a diameter more than 2cm.
	Open forest land	Land composed of arbor species with a canopy cover between 10%-19%.
	Shrub land	Land composed of shrubs and/or bushes with a canopy cover more than 30%, of which main purpose is to cultivate shrubs or used for protection.
	Unestablished afforestation land	Afforestation land that has not been closed but is promising for forest during 3-5 years.
	Nursery land	Land permanently used for cultivating tree seedlings and saplings.
	Cut-over & fired-over forest land	Forest land with a canopy cover less than 10% and within 3 years after clear cut or fired.
	Forest suitable land	Land planned for forestry development by county-level or higher level government.
	Non-forestry land	All land not classified into forestry land above, includes cultivated land, grassland, water area, unused land and construction land.

2006	National class	Definition
	Arbor forest	Forest land composed of arbor species with a canopy cover of more than 20%.
	Bamboo forest	Forest land composed of bamboo species with a diameter more than 2cm.
	Open forest land	Land composed of arbor species with a canopy cover between 10%-19%.
	Shrub land	Land composed of shrubs and/or bushes with a canopy cover more than 30%, of which main purpose is to cultivate shrubs or used for protection.

	Unestablished afforestation land	Afforestation land that has not been closed but is promising for forest during 3-5 years.
	Nursery land	Land permanently used for cultivating tree seedlings and saplings.
	Cut-over & fired-over forest land	Forest land with a canopy cover less than 10% and within 3 years after clear cut or fire.
	Forest suitable land	Land planned for forestry development by county-level or higher level government.
	Non-forestry land	All land not classified into forestry land above, includes cultivated land, grassland, water area, unused land and construction land.
	Economic forest	Forest land composed of economic species, of which main purpose is to produce fruits, cooking oil, drinks, spices, industrial raw materials and medicinal materials.

2011	National class	Definition
	Arbor forest	Forest land composed of arbor species with a canopy cover of more than 20%.
	Bamboo forest	Forest land composed of bamboo species with a diameter more than 2cm.
	Open forest land	Land composed of arbor species with a canopy cover between 10%-19%.
	Shrub land	Land composed of shrubs and/or bushes with a canopy cover more than 30%, of which main purpose is to cultivate shrubs or used for protection.
	Unestablished afforestation land	Afforestation land that has not been closed but is promising for forest during 3-5 years.
	Nursery land	Land permanently used for cultivating tree seedlings and saplings.
	Cut-over & fired-over forest land	Forest land with a canopy cover less than 10% and within 3 years after clear cut or fire.
	Forest suitable land	Land planned for forestry development by county-level or higher level government.
	Non-forestry land	All land not classified into forestry land above, includes cultivated land, grassland, water area, unused land and construction land.
	Economic forest	Forest land composed of economic species, of which main purpose is to produce fruits, cooking oil, drinks, spices, industrial raw materials and medicinal materials.

2016	National class	Definition
	Arbor forest	Forest land composed of arbor species with a canopy cover of more than 20%.
	Bamboo forest	Forest land composed of bamboo species with a diameter more than 2cm.

	Open forest land	Land composed of arbor species with a canopy cover between 10%-19%.
	Shrub land	Land composed of shrubs and/or bushes with a canopy cover more than 30%, of which main purpose is to cultivate shrubs or used for protection.
	Unestablished afforestation land	Afforestation land that has not been closed but is promising for forest during 3-5 years.
	Nursery land	Land permanently used for cultivating tree seedlings and saplings.
	Cut-over & fired-over forest land	Forest land with a canopy cover less than 10% and within 3 years after clear cut or fire.
	Forest suitable land	Land planned for forestry development by county-level or higher level government.
	Non-forestry land	All land not classified into forestry land above, includes cultivated land, grassland, water area, unused land and construction land.
	Economic forest	Forest land composed of economic species, of which main purpose is to produce fruits, cooking oil, drinks, spices, industrial raw materials and medicinal materials.

Original data and reclassification

	Classifications and definitions		FRA classes		
	Class	Area (1000 ha)	Forest	Other wooded land	Other land
1986	Arbor forest	107 248.80	100.00 %	0.00 %	0.00 %
	Economic forest	13 743.80	73.00 %	0.00 %	27.00 %
	Bamboo forest	3 660.20	100.00 %	0.00 %	0.00 %
	Open forest land	19 636.50	100.00 %	0.00 %	0.00 %
	Shrub land	28 116.00	0.00 %	100.00 %	0.00 %
	Unestablished afforestation land	7 288.10	100.00 %	0.00 %	0.00 %
	Nursery land	184.50	100.00 %	0.00 %	0.00 %
	Cut-over & fired-over forest land	4 430.60	100.00 %	0.00 %	0.00 %
	Forest suitable land	82 181.60	0.00 %	100.00 %	0.00 %
	Non-forestry land	693 509.90	0.00 %	0.00 %	100.00 %
	Total	960 000.00	152 481.67	110 297.60	697 220.73

1991	Classifications and definitions		FRA classes		
	Class	Area (1000 ha)	Forest	Other wooded land	Other land
	Arbor forest	113 700.00	100.00 %	0.00 %	0.00 %
	Economic forest	16 098.80	73.00 %	0.00 %	27.00 %
	Bamboo forest	3 904.70	100.00 %	0.00 %	0.00 %
	Open forest land	18 025.70	100.00 %	0.00 %	0.00 %
	Shrub land	29 706.30	0.00 %	100.00 %	0.00 %
	Unestablished afforestation land	7 138.30	100.00 %	0.00 %	0.00 %
	Nursery land	114.90	100.00 %	0.00 %	0.00 %
	Cut-over & fired-over forest land	3 669.60	100.00 %	0.00 %	0.00 %
	Forest suitable land	69 591.40	0.00 %	100.00 %	0.00 %
	Non-forestry land	698 050.30	0.00 %	0.00 %	100.00 %
	Total	960 000.00	158 305.32	99 297.70	702 396.98

1996	Classifications and definitions		FRA classes		
	Class	Area (1000 ha)	Forest	Other wooded land	Other land
	Arbor forest	134 355.70	100.00 %	0.00 %	0.00 %
	Economic forest	20 222.10	73.00 %	0.00 %	27.00 %
	Bamboo forest	4 363.10	100.00 %	0.00 %	0.00 %
	Open forest land	7 195.00	100.00 %	0.00 %	0.00 %
	Shrub land	34 445.70	0.00 %	100.00 %	0.00 %
	Unestablished afforestation land	4 615.10	100.00 %	0.00 %	0.00 %
	Nursery land	122.50	100.00 %	0.00 %	0.00 %
	Cut-over & fired-over forest land	3 106.80	100.00 %	0.00 %	0.00 %
	Forest suitable land	64 394.60	0.00 %	100.00 %	0.00 %

	Non-forestry land	687 179.40	0.00 %	0.00 %	100.00 %
	Total	960 000.00	168 520.33	98 840.30	692 639.37

2001	Classifications and definitions		FRA classes		
	Class	Area (1000 ha)	Forest	Other wooded land	Other land
	Arbor forest	144 736.80	100.00 %	0.00 %	0.00 %
	Economic forest	21 390.00	73.00 %	0.00 %	27.00 %
	Bamboo forest	4 994.90	100.00 %	0.00 %	0.00 %
	Open forest land	5 999.60	100.00 %	0.00 %	0.00 %
	Shrub land	45 296.80	0.00 %	100.00 %	0.00 %
	Unestablished afforestation land	4 893.60	100.00 %	0.00 %	0.00 %
	Nursery land	270.90	100.00 %	0.00 %	0.00 %
	Cut-over & fired-over forest land	2 610.10	100.00 %	0.00 %	0.00 %
	Forest suitable land	54 713.10	0.00 %	100.00 %	0.00 %
	Non-forestry land	675 094.20	0.00 %	0.00 %	100.00 %
	Total	960 000.00	179 120.60	100 009.90	680 869.50

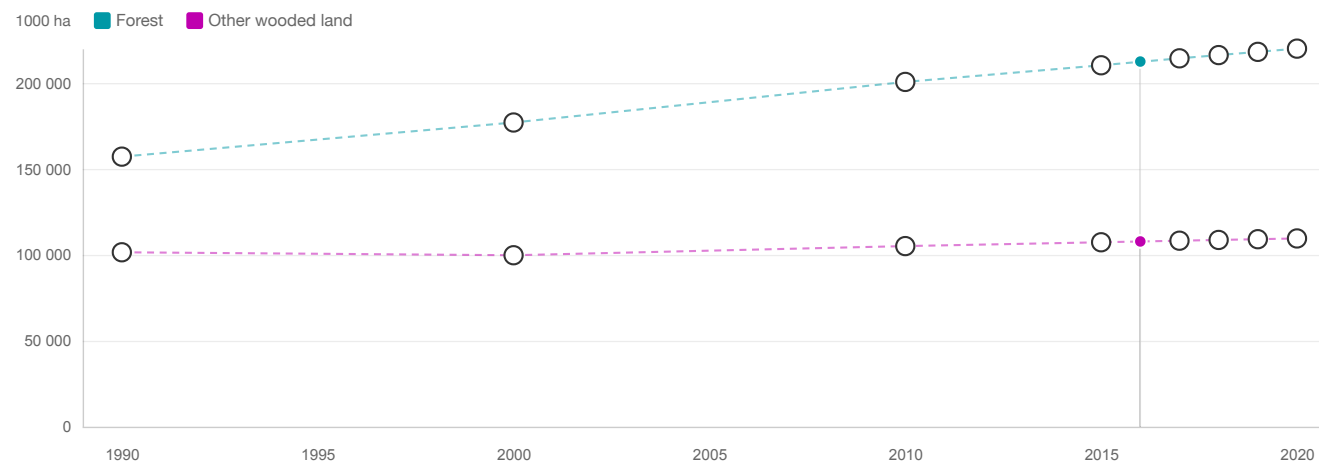
2006	Classifications and definitions		FRA classes		
	Class	Area (1000 ha)	Forest	Other wooded land	Other land
	Arbor forest	157 559.80	100.00 %	0.00 %	0.00 %
	Bamboo forest	5 533.30	100.00 %	0.00 %	0.00 %
	Open forest land	4 822.20	100.00 %	0.00 %	0.00 %
	Shrub land	53 653.40	0.00 %	100.00 %	0.00 %
	Unestablished afforestation land	10 461.80	100.00 %	0.00 %	0.00 %
	Nursery land	454.00	100.00 %	0.00 %	0.00 %

	Cut-over & fired-over forest land	2 794.30	100.00 %	0.00 %	0.00 %
	Forest suitable land	50 215.30	0.00 %	100.00 %	0.00 %
	Non-forestry land	654 095.90	0.00 %	0.00 %	100.00 %
	Economic forest	20 410.00	73.00 %	0.00 %	27.00 %
	Total	960 000.00	196 524.70	103 868.70	659 606.60

2011	Classifications and definitions		FRA classes		
	Class	Area (1000 ha)	Forest	Other wooded land	Other land
	Arbor forest	166 579.40	100.00 %	0.00 %	0.00 %
	Bamboo forest	6 158.60	100.00 %	0.00 %	0.00 %
	Open forest land	4 006.80	100.00 %	0.00 %	0.00 %
	Shrub land	55 902.20	0.00 %	100.00 %	0.00 %
	Unestablished afforestation land	6 502.60	100.00 %	0.00 %	0.00 %
	Nursery land	506.40	100.00 %	0.00 %	0.00 %
	Cut-over & fired-over forest land	2 865.40	100.00 %	0.00 %	0.00 %
	Forest suitable land	49 503.40	0.00 %	100.00 %	0.00 %
	Non-forestry land	647 410.00	0.00 %	0.00 %	100.00 %
	Economic forest	20 565.20	73.00 %	0.00 %	27.00 %
	Total	960 000.00	201 631.80	105 405.60	652 962.60

2016	Classifications and definitions		FRA classes		
	Class	Area (1000 ha)	Forest	Other wooded land	Other land
	Arbor forest	177 100.50	100.00 %	0.00 %	0.00 %
	Bamboo forest	6 524.10	100.00 %	0.00 %	0.00 %
	Open forest land	3 421.80	100.00 %	0.00 %	0.00 %
	Shrub land	57 822.90	0.00 %	100.00 %	0.00 %

	Unestablished afforestation land	6 991.40	100.00 %	0.00 %	0.00 %
	Nursery land	719.80	100.00 %	0.00 %	0.00 %
	Cut-over & fired-over forest land	2 424.90	100.00 %	0.00 %	0.00 %
	Forest suitable land	49 977.90	0.00 %	100.00 %	0.00 %
	Non-forestry land	634 088.80	0.00 %	0.00 %	100.00 %
	Economic forest	20 927.90	73.00 %	0.00 %	27.00 %
	Total	960 000.00	212 459.87	107 800.80	639 739.33



FRA categories	Area (1000 ha)								
	1990	2000	2010	2015	2016	2017	2018	2019	2020
Forest (a)	157 140.59	177 000.55	200 610.38	210 294.25	212 459.87	214 339.47	216 219.04	218 098.61	219 978.18
Other wooded land (a)	101 497.68	99 775.98	105 098.22	107 321.76	107 800.80	108 236.93	108 673.05	109 109.18	109 545.30
Other land (c-a-b)	683 831.73	665 693.47	636 761.40	624 853.99	622 209.33	619 893.60	617 577.91	615 262.21	612 946.52
Total land area (c)	942 470.00	942 470.00	942 470.00	942 470.00	942 470.00	942 470.00	942 470.00	942 470.00	942 470.00

The FAOSTAT land area figure for the year 2015 is used for all reference years

Climatic domain	% of forest area 2015	Override value
Boreal	7.00	
Temperate	22.00	
Sub-tropical	67.00	
Tropical	4.00	

Comments

The forest area and other wooded land area after 2016 are predicted using the mean of average annual increments of the last 5 and 10 years. The other land area is equal to total land area minus the forest and other woodland areas. Total land area is 9.6 million km² according to China's published data.

1b Forest characteristics

National Data

Data sources + type of data source eg NFI, etc

No	References to sources of information	Year(s)	Additional comments
1	State Forestry Administration, 1990, China Forest Resources Statistics (1984-1988)	1986	Data of 3 rd National Forest Inventory in China
2	State Forestry Administration, 1995, China Forest Resources Statistics (1989-1993)	1991	Data of 4 th National Forest Inventory in China
3	State Forestry Administration, 2000, China Forest Resources Statistics (1994-1998)	1996	Data of 5 th National Forest Inventory in China
4	State Forestry Administration, 2005, China Forest Resources Statistics (1999-2003)	2001	Data of 6 th National Forest Inventory in China
5	State Forestry Administration, 2010, China Forest Resources Statistics (2004-2008)	2006	Data of 7 th National Forest Inventory in China
6	State Forestry Administration, 2014, China Forest Resources Statistics (2009-2013)	2011	Data of 8 th National Forest Inventory in China
7	National Forestry and Grassland Administration, 2019, China Forest Resources Statistics (2014-2018)	2016	Data of 9 th National Forest Inventory in China
8	State Forestry Administration, 2015, FRA2020 Country Report -China.	1990-2010	Assessment results for reporting years

National classification and definitions

Origins	Definitions
Natural forest	Forests oriented from natural regeneration, or natural regeneration under artificial promotion.
Planted forest	Forests oriented from seedling planting, point seeding, or aerial seeding, including the coppice forest grown up from cut-over plantation.

Original data

1b.4.1 1984-2003 period

National classes	Area (1000ha)			
	1986 (1984-1988)	1991 (1989-1993)	1996 (1994-1998)	2001 (1999-2003)
Natural forest	93219.0	99029.3	111851.4	117441.8
Planted forest	27723.0	30327.5	41629.5	47924.4
Open forest land	19636.5	18025.7	7195.0	5999.6
Unestablished afforestation land	7288.1	7138.3	4615.1	4893.6
Nursery land	184.5	114.9	122.5	270.9
Cut-over & fired-over forest land	4430.6	3669.6	3106.8	2610.1

1b.4.2 2004-2018 period

National classes	Area (1000ha)		
	2006 (2004-2008)	2011 (2009-2013)	2016 (2014-2018)

Natural forest	121392.1	123546.8	128831.8
Planted forest	56600.3	64203.8	70070.2
Natural open forest land	3673.1	2869.1	2412.8
Planted open forest land	1149.1	1137.7	1009.0
Unestablished afforestation land	10461.8	6502.6	6991.4
Nursery land	454.0	506.4	719.8
Cut-over & fired-over forest land	2794.3	2865.4	2424.9

Analysis and processing of national data

Estimation and forecasting

1. Estimation of planted forest by categories

FRA2020 categories	Area (1000ha)						
	1986	1991	1996	2001	2006	2011	2016
Planted forest	42909.12	44462.70	49935.10	56054.04	70341.78	74069.74	80245.34
Plantation forest	34327.30	35570.16	37451.33	39237.83	45018.74	41479.05	43332.48
...of which introduced species	3432.73	4624.12	5992.21	7455.19	9904.12	10369.76	12133.10
Other plated forest	8581.82	8892.54	12483.78	16816.21	25323.04	32590.69	36912.86

Note: According to statistical analysis and expert estimation, the proportions of plantation forest to planted forest in the seven years are 80%, 80%, 75%, 70%, 64%, 56%, and 54%, respectively; and the proportions of introduced species to plantation forest are 10%, 13%, 16%, 19%, 22%, 25%, and 28%, respectively.

2. Estimation and forecasting by categories

FRA2020 categories	Area (1000 ha)										
	1990	1995	2000	2005	2010	2015	2016	2017	2018	2019	2020
Naturally regenerating forest	112988.60	122170.29	127286.23	131284.06	132214.56	132981.39	133748.22	134515.05	135281.88	112988.60	122170.29
Planted forest	44151.98	54830.25	73324.15	79010.22	80245.34	81358.08	82470.82	83583.56	84696.30	44151.98	54830.25
Plantation forest	35321.58	38880.53	42186.99	42961.80	43332.48	43933.36	44534.24	45135.12	45736.00	35321.58	38880.53
...of which introduced species	4385.84	7162.59	10276.64	11780.43	12133.10	13619.34	13805.61	13991.89	14178.16	4385.84	7162.59
Other planted forest	8830.40	15949.72	31137.16	36048.42	36912.86	37424.72	37936.58	38448.44	38960.30	8830.40	15949.72

Note: 1) The naturally regenerating forest area and planted forest area after 2016 are predicted using the mean of average annual increments of the last 5 and 10 years; 2) The area proportions of both plantation forest to planted forest and introduced species to plantation forest after 2016 are the same as in 2016, that is, 54% and 28%, respectively

Reclassification into FRA 2020 categories

1. Reclassification matrix

1.1 Data of four periods between 1984-2003

Classifications and definitions	FRA classes	
National classes	Naturally regenerating forest	Planted forest
Natural forest	100%	0%
Planted forest	0%	100%
Open forest land	72%	28%
Unestablished afforestation land	0%	100%
Nursery land	0%	100%
Cut-over forest land	50%	50%
Fired-over forest land	50%	50%

Note: The above reclassification matrix is based on the parameters of FRA2015.

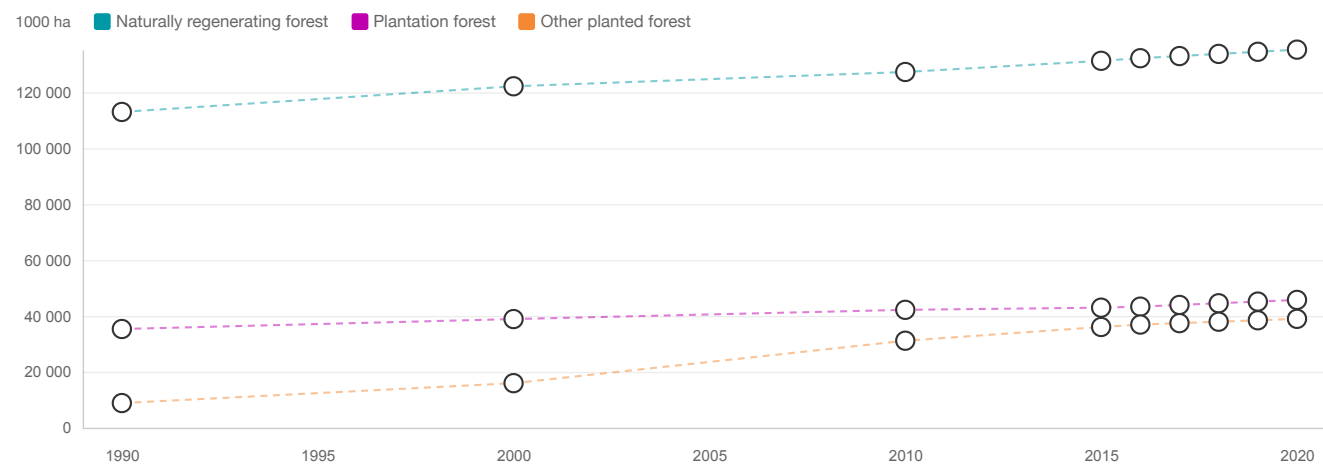
1.2 Data of three periods between 2004-2018

Classifications and definitions	FRA classes	
National classes	Naturally regenerating forest	Planted forest
Natural forest	100%	0%
Planted forest	0%	100%
Natural open forest land	100%	0%
Planted open forest land	0%	100%
Unestablished afforestation land	0%	100%
Nursery land	0%	100%
Cut-over & fired-over forest land	40%	60%

Note: The percentages of cut-over & fired-over forest land reclassified as naturally regenerating forest and planted forest are based on the area proportion of natural forest and artificial forest converted from cut-over & fired-over forest land during the seventh, eighth and ninth NFIs.

2. Reclassification results

FRA2020 categories	Area (1000 ha)						
	1986	1991	1996	2001	2006	2011	2016
Naturally regenerating forest	109572.58	113842.60	118585.20	123066.56	126182.92	127562.06	132214.56
Planted forest	42909.12	44462.70	49935.10	56054.04	70341.78	74069.74	80245.34



FRA categories	Forest area (1000 ha)								
	1990	2000	2010	2015	2016	2017	2018	2019	2020
Naturally regenerating forest (a)	112 988.60	122 170.29	127 286.23	131 284.06	132 214.56	132 981.39	133 748.22	134 515.05	135 281.88
Planted forest (b)	44 151.98	54 830.25	73 324.15	79 010.22	80 245.34	81 358.08	82 470.82	83 583.56	84 696.30
Plantation forest	35 321.58	38 880.53	42 186.99	42 961.80	43 332.48	43 933.36	44 534.24	45 135.12	45 736.00
...of which introduced species	4 385.84	7 162.59	10 276.64	11 780.43	12 133.10	13 619.34	13 805.61	13 991.89	14 178.16
Other planted forest	8 830.40	15 949.72	31 137.16	36 048.42	36 912.86	37 424.72	37 936.58	38 448.44	38 960.30
Total (a+b)	157 140.58	177 000.54	200 610.38	210 294.28	212 459.90	214 339.47	216 219.04	218 098.61	219 978.18
Total forest area	157 140.59	177 000.55	200 610.38	210 294.25	212 459.87	214 339.47	216 219.04	218 098.61	219 978.18

Comments

1c Primary forest and special forest categories

National Data

Data sources + type of data source eg NFI, etc

No	References to sources of information	Year(s)	Additional comments
1	State Forestry Administration, 1990, China Forest Resources Statistics (1984-1988)	1986	Data of 3 rd National Forest Inventory in China
2	State Forestry Administration, 1995, China Forest Resources Statistics (1989-1993)	1991	Data of 4 th National Forest Inventory in China
3	State Forestry Administration, 2000, China Forest Resources Statistics (1994-1998)	1996	Data of 5 th National Forest Inventory in China
4	State Forestry Administration, 2005, China Forest Resources Statistics (1999-2003)	2001	Data of 6 th National Forest Inventory in China
5	State Forestry Administration, 2010, China Forest Resources Statistics (2004-2008)	2006	Data of 7 th National Forest Inventory in China
6	State Forestry Administration, 2014, China Forest Resources Statistics (2009-2013)	2011	Data of 8 th National Forest Inventory in China
7	National Forestry and Grassland Administration, 2019, China Forest Resources Statistics (2014-2018)	2016	Data of 9 th National Forest Inventory in China
8	State Forestry Administration, 2014, FRA2015 Country Report -China.	1990-2010	Assessment results for reporting years
9	Jia MingMing. 2014. Remote sensing-based analysis on changes of Mangroves in China in 1973-2013.	1973-2013	Analysis of Mangrove areas in China based on remote sensing
10	State Forestry Administration. 2002. Inventory Report on Mangroves in China.	2001	Inventory of Mangrove areas in China based on field survey

National classification and definitions

Categories	Definitions
Bamboo forest	Forest land composed of bamboo species with a diameter more than 2cm, and with a canopy cover of more than 20%.
Unestablished afforestation land	Afforestation land that has not been closed but is promising for forest during 3-5 years.
Cut-over forest land	Forest land with a canopy cover less than 10% and within 3 years after clear cut.
Fired-over forest land	Forest land with a canopy cover less than 10% and within 3 years after fired.
Natural arbor forest	Naturally regenerated forest land composed of arbor species, with a canopy cover of more than 20%.
...of which in <i>naturalness I</i>	Natural forest that is primitive or has little human influence and is almost in primitive state.
<i>naturalness II</i>	Natural forest that has obvious human disturbance, or secondary forest type in late ecological succession, and the dominant tree species are those in climax community.
<i>naturalness III</i>	Natural forest that has large human disturbance, and in late ecological succession, and the dominant tree species are pioneer species but the climax species are clearly visible.
<i>naturalness IV</i>	Natural forest that has very large human disturbance, and in retrogressive succession, or in extremely deformed secondary forest stage.
<i>naturalness V</i>	Natural forest that has very large and continuous human disturbance, and in late retrogressive succession, where it is difficult to return to positive succession.

Original data

1 Mangroves

Years	1973	1980	1990	2000	2001*	2010	2013
Mangroves (ha)	48750	22692	20391	18587	22351	20776	32834

Note: The data in 2001 marked with “*” was from field inventory, and the others were from remote sensing based survey, where the interpolated estimate in 2001 is 18806 ha.

2 Bamboos and other categories

(1) 1984-2003 period

National categories	Area (1000ha)			
	1986 (1984-1988)	1991 (1989-1993)	1996 (1994-1998)	2001 (1999-2003)
Bamboos	3660.2	3904.7	4363.1	4994.9
Unestablished afforestation land	7288.1	7138.3	4615.1	4893.6
Cut-over forest land	3096.0	2756.8	2506.0	1802.3
Fired-over forest land	1334.6	912.8	600.8	807.8
Arbor forest lower than 1.3m	1072.5	1137.0	1343.6	1750.0

(2) 2004-2018 period

National categories		Area (1000ha)		
		2006 (2004-2008)	2011 (2009-2013)	2016 (2014-2018)
Bamboos		5533.3	6158.6	6524.1
Unestablished afforestation land		10461.8	6502.6	6991.4
Cut-over & fired-over forest land		2794.3	2865.4	2424.9
Arbor forest lower than 1.3m		2018.8	1354.9	728.6
Natural arbor forest	Naturalness I	7796.3	6709.7	8569.9
	Naturalness II	23459.7	22171.6	28096.1
	Naturalness III	26456.4	30806.3	31643.8
	Naturalness IV	43276.7	42558.0	42916.6
	Naturalness V	14602.1	15288.3	11535.4

Analysis and processing of national data

Estimation and forecasting

1 Mangroves

Year	1990	2000	2010	2015	2020
Mangroves (1000ha)	24.469	22.304	24.931	29.296	31.777

Note: The areas in 1990, 2000, and 2010 from remote sensing (RS)-based survey were corrected using the ratio (1.2) of field inventory area to RS-based survey area in 2001. Considering that the change during 2010-2013 is large, the data in 2015 and 2020 were predicted using regression model based on the data from 2001 to 2013.

2 Bamboos, temporarily unstocked forest and primary forest

FRA2020 categories	Area (1000 ha)						
	1990	1995	2000	2005	2010	2015	2020
Bamboos	3856	4271	4869	5426	6034	6451	6817
Temporarily unstocked forest	12114	9641	9216	14071	11633	10261	9683
Primary forest	11646	11505	11453	11453	11453	11453	11453

Note: The areas of bamboos and temporarily unstocked forest in 2020 were predicted using linear extrapolation from 2011 to 2016; The primary forest areas during 2000-2020 were estimated using the average value of three periods from 2006-2016, and the primary forest area in 1990 was the same data as in FRA2015, and the primary forest area in 1995 was the average value of those in 1990 and 2000.

Reclassification into FRA 2020 categories

1 Reclassification matrix

(1) 1984-2003 period

Classifications and definitions	FRA categories		
National classes	Bamboos	Temporarily unstocked and /or recently regenerated forest	Primary forest
Bamboos	100%	0%	0%
Unestablished afforestation land	0%	100%	0%
Cut-over forest land	0%	100%	0%
Fired-over forest land	0%	100%	0%
Arbor forest lower than 1.3m	0%	100%	0%

(2) 2004-2018 period

Classifications and definitions		FRA categories		
National classes		Bamboos	Temporarily unstocked and /or recently regenerated forest	Primary forest
Bamboos		100%	0%	0%
Unestablished afforestation land		0%	100%	0%
Cut-over & fired-over forest land		0%	100%	0%
Arbor forest lower than 1.3m		0%	100%	0%
Natural arbor forest	Naturalness I	0%	0%	85%
	Naturalness II	0%	0%	20%
	Naturalness III	0%	0%	0%
	Naturalness IV	0%	0%	0%
	Naturalness V	0%	0%	0%

2 Reclassification results

FRA2020 categories	Area (1000 ha)						
	1986	1991	1996	2001	2006	2011	2016
Bamboos	3660.2	3904.7	4363.1	4994.9	5533.3	6158.6	6524.1
Temporarily unstocked forest	12791.2	11944.9	9065.5	9253.7	15274.9	10722.9	10144.9
Primary forest					11318.8	10137.6	12903.6

FRA categories	Area (1000 ha)				
	1990	2000	2010	2015	2020
Primary forest	11 646.00	11 453.00	11 453.33	11 453.33	11 453.33
Temporarily unstocked and/or recently regenerated	12 114.16	9 216.06	11 633.30	10 260.50	9 682.50
Bamboos	3 855.80	4 868.54	6 033.54	6 451.00	6 816.50
Mangroves	24.47	22.30	24.93	29.30	31.78
Rubber wood					

Comments

1d Annual forest expansion, deforestation and net change

National Data

Data sources + type of data source eg NFI, etc

No	References to sources of information	Year(s)	Additional comments
1	State Forestry Administration, 2000, China Forest Resources Statistics (1994-1998)	1996	Data of 5 th National Forest Inventory in China
2	State Forestry Administration, 2005, China Forest Resources Statistics (1999-2003)	2001	Data of 6 th National Forest Inventory in China
3	State Forestry Administration, 2010, China Forest Resources Statistics (2004-2008)	2006	Data of 7 th National Forest Inventory in China
4	State Forestry Administration, 2014, China Forest Resources Statistics (2009-2013)	2011	Data of 8 th National Forest Inventory in China
5	National Forestry and Grassland Administration, 2019, China Forest Resources Statistics (2014-2019)	2016	Data of 9 th National Forest Inventory in China

National classification and definitions

Same as the FRA classification.

Original data

Categories	Area (1000 ha/year)			
	1996-2001	2001-2006	2006-2011	2011-2016
Forest expansion	2421.3	2960.1	2378.9	2238.7
...of which afforestation	1269.8	1693.8	1497.3	1151.8
...of which natural expansion	1151.5	1266.3	881.6	1086.9

Analysis and processing of national data

Estimation and forecasting

Categories	Area (1000ha)					
	1990-1995	1995-2000	2000-2005	2005-2010	2010-2015	2015-2020
Forest expansion	2313.54	2313.54	2852.34	2495.14	2266.74	2069.99
...of which afforestation	1185.00	1185.00	1609.00	1536.60	1220.90	1067.80
...of which natural expansion	1128.54	1128.54	1243.34	958.54	1045.84	1002.19

Note: The data during 2015-2020 was predicted using linear extrapolation from 2000 to 2015 for afforestation, and using the average value from 2005 to 2015 for natural expansion; the data during 1995-2000 was predicted using linear extrapolation from 1996 to 2006; and the data during 1990-1995 was estimated as the same during 1995-2000.

Reclassification into FRA 2020 categories

Reclassification is not necessary.

FRA categories	Area (1000 ha/year)			
	1990-2000	2000-2010	2010-2015	2015-2020
Forest expansion (a)	2 313.54	2 673.74	2 266.74	2 069.99
...of which afforestation	1 185.00	1 572.80	1 220.90	1 067.80
...of which natural expansion	1 128.54	1 100.94	1 045.84	1 002.19
Deforestation (b)	327.54	312.76	329.97	133.20
Forest area net change (a-b)	1 986.00	2 360.98	1 936.77	1 936.79

Comments

1e Annual reforestation

National Data

Data sources + type of data source eg NFI, etc

No	References to sources of information	Year(s)	Additional comments
1	State Forestry Administration, 1991-2018, China Forestry Statistics Yearbook	1990-2017	Area of reforestation

National classification and definitions

Same as the FRA classification.

Original data

Years	Reforestation (1000ha)	Years	Reforestation (1000ha)
1990	671.50	2004	319.31
1991	664.10	2005	407.55
1992	673.60	2006	408.24
1993	739.20	2007	390.91
1994	722.70	2008	424.00
1995	751.00	2009	344.25
1996	794.80	2010	306.71
1997	798.40	2011	326.64
1998	806.30	2012	305.07
1999	1042.83	2013	303.09
2000	919.80	2014	292.45
2001	515.29	2015	299.58
2002	379.00	2016	272.80
2003	285.99	2017	305.44

Analysis and processing of national data

Estimation and forecasting

Years	2018	2019	2020
Reforestation (1000ha)	270.28	261.83	253.37

Note: The data during 2018-2020 was predicted using linear extrapolation from 2001 to 2017.

Reclassification into FRA 2020 categories

Reclassification is not necessary.

FRA categories	Area (1000 ha/year)			
	1990-2000	2000-2010	2010-2015	2015-2020
Reforestation	791.27	378.13	305.37	272.74

Comments

1f Other land with tree cover

National Data

Data sources + type of data source eg NFI, etc

No	References to sources of information	Year(s)	Additional comments
1	State Forestry Administration, 1990, China Forest Resources Statistics (1984-1988)	1986	Data of 3 rd National Forest Inventory in China
2	State Forestry Administration, 1995, China Forest Resources Statistics (1989-1993)	1991	Data of 4 th National Forest Inventory in China
3	State Forestry Administration, 2000, China Forest Resources Statistics (1994-1998)	1996	Data of 5 th National Forest Inventory in China
4	State Forestry Administration, 2005, China Forest Resources Statistics (1999-2003)	2001	Data of 6 th National Forest Inventory in China
5	State Forestry Administration, 2010, China Forest Resources Statistics (2004-2008)	2006	Data of 7 th National Forest Inventory in China
6	State Forestry Administration, 2014, China Forest Resources Statistics (2009-2013)	2011	Data of 8 th National Forest Inventory in China
7	National Forestry and Grassland Administration, 2019, China Forest Resources Statistics (2014-2018)	2016	Data of 9 th National Forest Inventory in China

National classification and definitions

Same as the FRA classification.

Original data

Years	Area (1000ha)	
	Palms	Tree orchards
1986 (1984-1988)	ID	2738.7
1991 (1989-1993)	ID	4249.1
1996 (1994-1998)	ID	7350.2
2001 (1999-2003)	140.0	8689.0
2006 (2004-2008)	192.0	8792.8
2011 (2009-2013)	413.0	8896.9
2016 (2014-2018)	188.3	9842.9

Note: Palms include betel nut, coconut, oil palm and palm; Tree orchards include tree & shrub orchards.

Analysis and processing of national data

Estimation and forecasting

FRA2020 Categories	Area (1000ha)						
	1990	1995	2000	2005	2010	2015	2020
Palms	97.82	134.41	171.00	207.59	244.18	280.77	317.36
Tree orchards	3947.02	6729.98	8421.24	8772.04	8876.08	9653.70	10178.75

Reclassification into FRA 2020 categories

Reclassification is not necessary.

FRA categories	Area (1000 ha)				
	1990	2000	2010	2015	2020
Palms (a)	97.82	171.00	244.18	280.77	317.36
Tree orchards (b)	3 947.02	8 421.24	8 876.08	9 653.70	10 178.75
Agroforestry (c)					
Trees in urban settings (d)					
Other (specify in comments) (e)					
Total (a+b+c+d+e)	4 044.84	8 592.24	9 120.26	9 934.47	10 496.11
Other land area	683 831.73	665 693.47	636 761.40	624 853.99	612 946.52

Comments

2 Forest growing stock, biomass and carbon

2a Growing stock

National Data

Data sources + type of data source eg NFI, etc

No	References to sources of information	Year(s)	Additional comments
1	State Forestry Administration, 1990, China Forest Resources Statistics (1984-1988)	1986	Data of 3 rd National Forest Inventory in China
2	State Forestry Administration, 1995, China Forest Resources Statistics (1989-1993)	1991	Data of 4 th National Forest Inventory in China
3	State Forestry Administration, 2000, China Forest Resources Statistics (1994-1998)	1996	Data of 5 th National Forest Inventory in China
4	State Forestry Administration, 2005, China Forest Resources Statistics (1999-2003)	2001	Data of 6 th National Forest Inventory in China
5	State Forestry Administration, 2010, China Forest Resources Statistics (2004-2008)	2006	Data of 7 th National Forest Inventory in China
6	State Forestry Administration, 2014, China Forest Resources Statistics (2009-2013)	2011	Data of 8 th National Forest Inventory in China
7	National Forestry and Grassland Administration, 2019, China Forest Resources Statistics (2014-2018)	2016	Data of 9 th National Forest Inventory in China
8	State Forestry Administration, 2014, FRA2015 Country Report –China	1990-2010	Assessment results for reporting years

National classification and definitions

Categories	Definitions
Growing stock	Volume over bark of all living trees with a minimum diameter of 5 cm at breast height.
...of which in arbor forest	Growing stock in arbor forest, which is classified into two types by origin, natural and planted.
...of which in open forest	Growing stock in open forest land, which is classified into two types by origin, natural and planted.
...of which for scattered trees	Growing stock except for those in arbor forest and open forest land, including the big old trees in young and middle-aged forests, excluding four-side trees.
...of which for four-side trees	Growing stock beside village, road, water, and field on non-forestry land.

Original data

Years	Growing stock (million m³)					
	Natural arbor forest	Planted arbor forest	Natural open forest land	Planted open forest land	Scattered trees	Four-side trees
1986 (1984-1988)	8582.1	559.0	513.3	32.5	694.5	191.0
1991 (1989-1993)	9395.6	741.1	510.6	34.3	771.4	332.1
1996 (1994-1998)	10207.6	1059.0	121.3	14.8	703.9	381.3
2001 (1999-2003)	10905.3	1550.6	114.3	13.9	710.9	323.2
2006 (2004-2008)	11714.3	2006.6	99.0	15.3	745.2	332.4
2011 (2009-2013)	12608.0	2529.3	89.5	16.2	789.1	400.7

2016 (2014-2018)	14108.1	3452.1	83.1	17.2	878.0	468.6
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Analysis and processing of national data

Estimation and forecasting

1 Estimation of planted forest by categories

FRA2020 categories	Growing stock (million m ³)						
	1986	1991	1996	2001	2006	2011	2016
Planted forest	591.5	775.4	1073.8	1564.5	2021.9	2545.5	3469.3
Plantation forest	325.3	426.5	569.1	782.3	950.3	1120.0	1491.8
Other plated forest	266.2	348.9	504.7	782.2	1071.6	1425.5	1977.5

Note: According to statistical analysis and expert estimation, the stock proportions of plantation forest to planted forest in the seven years are 55%, 55%, 53%, 50%, 47%, 44%, and 43%, respectively.

2 Estimating and forecasting by categories

FRA2020 categories	Growing stock (million m ³ over bark)										
	1990	1995	2000	2005	2010	2015	2016	2017	2018	2019	2020
Naturally regenerating forest	9744.04	10244.36	10881.46	11654.56	12520.66	13892.46	14191.20	14428.99	14666.78	14904.57	15142.36
Planted forest	738.62	1014.12	1466.36	1930.42	2440.78	3284.54	3469.30	3614.04	3758.78	3903.52	4048.26
Plantation forest	406.24	540.59	739.62	916.68	1086.07	1417.44	1491.80	1554.04	1616.28	1678.51	1740.75
Other plated forest	332.38	473.53	726.74	1013.74	1354.71	1867.10	1977.50	2060.00	2142.50	2225.01	2307.51
Other wooded land	756.02	717.40	709.50	738.34	780.32	860.22	878.00	891.28	904.56	917.84	931.12

Note: 1) The growing stocks of naturally regenerating forest, planted forest and other wooded land after 2016 were predicted using linear extrapolation from 2006 to 2016; 2) The plantation forest stock in 2020 was estimated as 43% of planted forest stock, and the other planted forest stocks are equal to planted forest stocks minus plantation forest stocks.

Reclassification into FRA 2020 categories

1 Reclassification matrix

Classifications and definitions	FRA categories		
National classes	Naturally regenerating forest	Planted forest	Other wooded land
Natural arbor forest	100%	0%	0%
Planted arbor forest	0%	100%	0%
Natural open forest land	100%	0%	0%
Planted open forest land	0%	100%	0%
Scattered trees	0%	0%	100%
Four-side trees	0%	0%	0%

Note: The difference between China's growing stock and FRA definition in the minimum diameter was not taken into consideration.

2 Reclassification results

FRA2020 categories	Growing stock (million m ³)						
	1986	1991	1996	2001	2006	2011	2016
Naturally regenerating forest	9095.4	9906.2	10328.9	11019.6	11813.3	12697.5	14191.2
Planted forest	591.5	775.4	1073.8	1564.5	2021.9	2545.5	3469.3
Other wooded land	694.5	771.4	703.9	710.9	745.2	789.1	878.0

FRA categories	Growing stock m³/ha (over bark)								
	1990	2000	2010	2015	2016	2017	2018	2019	2020
Naturally regenerating forest	86.24	89.07	98.37	105.82	107.33	108.50	109.66	110.80	111.93
Planted forest	16.73	26.74	33.29	41.57	43.23	44.42	45.58	46.70	47.80
...of which plantation forest	11.50	19.02	25.74	32.99	34.43	35.37	36.29	37.19	38.06
...of which other planted forest	37.64	45.56	43.51	51.79	53.57	55.04	56.48	57.87	59.23
Forest	66.71	69.76	74.58	81.68	83.12	84.18	85.22	86.24	87.24
Other wooded land	7.45	7.11	7.42	8.02	8.14	8.23	8.32	8.41	8.50

FRA categories	Total growing stock (million m³ over bark)								
	1990	2000	2010	2015	2016	2017	2018	2019	2020
Naturally regenerating forest	9 744.04	10 881.46	12 520.66	13 892.46	14 191.20	14 428.99	14 666.78	14 904.57	15 142.36
Planted forest	738.62	1 466.36	2 440.78	3 284.54	3 469.30	3 614.04	3 758.78	3 903.52	4 048.26
...of which plantation forest	406.24	739.62	1 086.07	1 417.44	1 491.80	1 554.04	1 616.28	1 678.51	1 740.75
...of which other planted forest	332.38	726.74	1 354.71	1 867.10	1 977.50	2 060.00	2 142.50	2 225.01	2 307.51
Forest	10 482.66	12 347.82	14 961.44	17 177.00	17 660.50	18 043.03	18 425.56	18 808.09	19 190.62
Other wooded land	756.02	709.50	780.32	860.22	878.00	891.28	904.56	917.84	931.12

Comments

We fill in the second table first, and then calculate the Growing stock per hectare in the first table, using the corresponding area figures in table 1b.

2b Growing stock composition

National Data

Data sources + type of data source eg NFI, etc

Same as in 2a. Additionally, the following book was referenced:

Pan Zhigang, You Yingtian, et al. Growing exotic trees in China. Beijing Science & Technology Press, 1994.

National classification and definitions

Same as the FRA classification.

Original data

Tree species	Growing stock in forest (million m ³ over bark)						
	1986	1991	1996	2001	2006	2011	2016
<i>Quercus</i>	1101.97	1208.69	1335.79	1321.40	1208.41	1277.56	1386.57
<i>Abies</i>	800.80	1074.98	1098.84	1194.70	1135.62	1166.05	1325.03
<i>Larix</i>	940.07	871.97	940.89	920.55	955.22	1001.26	1122.96
<i>Picea</i>	904.63	1126.71	1279.08	1036.15	1001.60	998.99	972.66
<i>Betula</i>	600.36	672.09	775.52	845.70	799.46	917.50	922.85
<i>Cunninghamia lanceolata</i>	268.52	342.65	473.57	734.82	734.09	726.02	852.02
<i>Populus</i>	246.21	281.79	359.30	425.97	549.39	591.00	702.18
<i>Pinus massoniana</i>	407.19	430.21	558.70	671.68	587.88	623.84	626.06
<i>Pinus yunnanensis</i>	253.81	240.12	279.16	518.23	468.72	502.28	501.01
<i>Pinus densata</i>	153.61	69.04	197.79	378.18	354.26	349.69	332.41
Remaining native tree species	3816.03	4096.38	3761.95	4096.19	5486.99	6402.87	8033.73
TOTAL volume of native tree species	9493.19	10414.61	11060.58	12143.57	13281.64	14557.06	16777.47
<i>Eucalyptus</i>	ID	ID	ID	19.37	45.81	160.36	215.63
<i>Pinus elliottii, etc</i>	ID	ID	ID	ID	55.91	48.87	66.81
<i>Robinia pseudoacacia</i>	ID	ID	ID	ID	ID	27.22	52.23
<i>Acacia mangium, etc</i>	ID	ID	ID	ID	7.57	12.19	14.63
<i>Casuarina</i>	ID	ID	ID	7.20	4.63	3.70	2.33

<i>equisetifolia</i>							
Remaining introduced tree species	193.71	266.99	342.12	413.95	439.65	433.60	531.40
TOTAL volume of introduced tree species	193.71	266.99	342.12	440.53	553.65	685.93	883.02

Note: The ranking orders of top ten native species and top five introduced species are according to volume in 2016. The proportion of introduced tree species volume to total forest volume is gradually increased from 2% in 1986 to 5% in 2016. Other introduced tree species include Japanese larch (*Larix kaempferi*), American poplar (*Populus × canadensis*), plane tree (*Platanus* spp.) and so on, but the data of their growing stocks can't be obtained, because there are no separate tree species codes for them in the NFIs of China.

Analysis and processing of national data

Estimation and forecasting

Tree species	Growing stock in forest (million m ³ over bark)						
	1990	1995	2000	2005	2010	2015	2020
<i>Quercus</i>	1187.34	1310.37	1324.28	1231.01	1263.73	1364.77	1457.83
<i>Abies</i>	1020.14	1094.07	1175.53	1147.44	1159.97	1293.23	1400.79
<i>Larix</i>	885.59	927.10	924.62	948.28	992.05	1098.62	1190.05
<i>Picea</i>	1082.29	1248.60	1084.74	1008.51	999.51	977.93	961.09
<i>Betula</i>	657.75	754.83	831.67	808.71	893.89	921.78	972.21
<i>Cunninghamia lanceolata</i>	327.82	447.39	682.57	734.24	727.63	826.82	899.19
<i>Populus</i>	274.67	343.80	412.63	524.71	582.68	679.94	763.29
<i>Pinus massoniana</i>	425.60	533.00	649.08	604.64	616.65	625.62	641.34
<i>Pinus yunnanensis</i>	242.86	271.35	470.42	478.62	495.57	501.26	513.92
<i>Pinus densata</i>	85.95	172.04	342.10	359.04	350.60	335.87	323.67
Remaining native tree species	4040.31	3828.83	4029.34	5208.83	6219.69	7707.55	9052.42
TOTAL volume of native tree species	10230.33	10931.39	11926.97	13054.03	14301.98	16333.39	18175.81
<i>Eucalyptus</i>	ID	ID	14.08	40.52	137.45	204.58	283.56
<i>Pinus elliotii</i> , etc	ID	ID	ID	57.32	50.27	63.22	71.16
<i>Robinia pseudoacacia</i>	ID	ID	ID	ID	22.22	47.23	73.13
<i>Acacia mangium</i> , etc	ID	ID	ID	6.65	11.27	14.14	17.46
<i>Casuarina</i>	ID	ID	7.72	5.14	3.88	2.60	1.41

<i>equisetifolia</i>							
Remaining introduced tree species	252.33	327.09	399.04	421.33	434.36	511.84	568.10
TOTAL volume of introduced tree species	252.33	327.09	420.85	530.95	659.46	843.61	1014.81

Reclassification into FRA 2020 categories

Reclassification is not necessary.

FRA categories	Scientific name	Common name	Growing stock in forest (million m³ over bark)				
			1990	2000	2010	2015	2020
Native tree species							
#1 Ranked in terms of volume	Quercus	Oak	1 187.34	1 324.28	1 263.73	1 364.77	1 457.83
#2 Ranked in terms of volume	Abies	Fir	1 020.14	1 175.53	1 159.97	1 293.23	1 400.79
#3 Ranked in terms of volume	Larix	Larch	885.59	924.62	992.05	1 098.62	1 190.05
#4 Ranked in terms of volume	Picea	Spruce	1 082.29	1 084.74	999.51	977.93	961.09
#5 Ranked in terms of volume	Betula	Birch	657.75	831.67	893.89	921.78	972.21
#6 Ranked in terms of volume	Cunninghamia lanceolata	China fir	327.82	682.57	727.63	826.82	899.19
#7 Ranked in terms of volume	Populus	Poplar	274.67	412.63	582.68	679.94	763.29
#8 Ranked in terms of volume	Pinus massoniana	Manson pine	425.60	649.08	616.65	625.62	641.34
#9 Ranked in terms of volume	Pinus yunnanensis	Yunan pine	242.86	470.42	495.57	501.26	513.92
#10 Ranked in terms of volume	Pinus densata	Alpine pine	85.95	342.10	350.60	335.87	323.67
Remaining native tree species			4 040.32	4 029.33	6 219.70	7 707.55	9 052.42
Total volume of native tree species			10 230.33	11 926.97	14 301.98	16 333.39	18 175.80
Introduced tree species							
#1 Ranked in terms of volume	Eucalyptus	Eucalypt		14.08	137.45	204.58	283.56

FRA categories	Scientific name	Common name	Growing stock in forest (million m³ over bark)				
			1990	2000	2010	2015	2020
Native tree species							
#2 Ranked in terms of volume	Pinus elliottii, etc	Foreign pine			50.27	63.22	71.16
#3 Ranked in terms of volume	Robinia pseudoacacia	Locust			22.22	47.23	73.13
#4 Ranked in terms of volume	Acacia mangium, etc	Acacia			11.27	14.14	17.46
#5 Ranked in terms of volume	Casuarina equisetifolia	Coast oak		7.72	3.88	2.60	1.41
Remaining introduced tree species			252.33	399.04	434.36	511.84	568.10
Total volume of introduced tree species			252.33	420.84	659.45	843.61	1 014.82
Total growing stock			10 482.66	12 347.81	14 961.43	17 177.00	19 190.62

Comments

The ranking orders of top ten native species and top five introduced species are according to volume in 2015. The proportion of introduced tree species volume to total forest volume is gradually increased from 2% in 1986 to 5% in 2015. The data of top 5 introduced tree species before 2000 are not available. Other introduced tree species include Japanese larch (*Larix kaempferi*), American poplar (*Populus × canadensis*), plane tree (*Platanus* spp.) and so on, but the data of their growing stocks can't be obtained, because there are no separate tree species codes for them in the NFIs of China.

2c Biomass stock

National Data

Data sources + type of data source eg NFI, etc

Same as in 2a.

National classification and definitions

Same as the FRA classification.

Original data

National categories	Forest Biomass (million tonnes)						
	1986	1991	1996	2001	2006	2011	2016
Above-ground biomass of arbor forest and open forest	5874.02	6476.67	6988.74	7692.74	8070.49	8890.67	10262.90
Below-ground biomass of arbor forest and open forest	2218.20	2475.71	2710.60	2946.53	3056.79	3328.74	3832.67
Above-ground biomass of bamboos	403.41	428.92	478.79	550.04	615.50	691.20	730.93
Below-ground biomass of bamboos	152.34	163.95	185.70	210.68	233.13	258.79	272.96
Total of above-ground biomass	6277.43	6905.59	7467.53	8242.78	8685.99	9581.86	10993.83
Total of below-ground biomass	2370.54	2639.66	2896.30	3157.21	3289.92	3587.53	4105.63
Dead wood	1350.17	1490.61	1605.40	1763.71	1921.68	2110.29	2434.30

Note: The biomass data in this table was calculated from the BEF method with the IPCC default values recommended in the FRA2020 Guidelines and Specifications.

National categories	Forest Biomass (million tons)						
	1986	1991	1996	2001	2006	2011	2016
Above-ground biomass of arbor forest and open forest	7350.30	8104.41	8745.17	9626.10	10098.79	11125.10	12842.21
Below-ground biomass of arbor forest and open forest	1854.12	2069.36	2265.70	2462.91	2555.07	2782.38	3203.60
Above-ground biomass of bamboos	153.98	163.72	182.76	209.95	234.94	263.83	279.00
Below-ground biomass of bamboos	79.79	85.87	97.26	110.35	122.11	135.55	142.97
Total of above-ground biomass	7504.28	8268.13	8927.93	9836.06	10333.73	11388.93	13121.20
Total of below-ground biomass	1933.91	2155.23	2362.96	2573.25	2677.17	2917.92	3346.57
Dead wood	1473.55	1627.74	1749.01	1919.86	2087.76	2292.56	2654.90

Note: The biomass data in 2016 in this table was calculated according to the ministerial standards on tree biomass models and related parameters to carbon accounting published in recent years, and the biomass data in other years were calculated using the proportions of two sets of biomass data in 2016 as correction factors based on the data in the table above. The data of dead wood were also calculated using the proportions in the table above

Analysis and processing of national data

Estimation and forecasting

Categories	Forest Biomass (million tons)								

	1990	2000	2010	2015	2016	2017	2018	2019	2020
Above-ground biomass	8115	9654	11178	12775	13121	13400	13679	13957	14236
Below-ground biomass	2111	2531	2870	3261	3347	3414	3480	3547	3614
Dead wood	1597	1886	2252	2582	2655	2655	2655	2655	2655

Note: The biomass data after 2016 was predicted using linear extrapolation from 2006 to 2016.

Reclassification into FRA 2020 categories

Reclassification is not necessary.

FRA categories	Forest biomass (tonnes/ha)								
	1990	2000	2010	2015	2016	2017	2018	2019	2020
Above-ground biomass	51.64	54.54	55.72	60.75	61.76	62.52	63.26	64.00	64.72
Below-ground biomass	13.43	14.30	14.31	15.51	15.75	15.93	16.10	16.27	16.43
Dead wood	10.16	10.65	11.22	12.28	12.50	12.39	12.28	12.17	12.07

Comments

2d Carbon stock

National Data

Data sources + type of data source eg NFI, etc

Same as in 2a.

National classification and definitions

Same as the FRA classification.

Original data

Categories	Forest Biomass (million tons)								
	1990	2000	2010	2015	2016	2017	2018	2019	2020
Above-ground biomass	8115	9654	11178	12775	13121	13400	13679	13957	14236
Below-ground biomass	2111	2531	2870	3261	3347	3414	3480	3547	3614
Dead wood	1597	1886	2252	2582	2655	2655	2655	2655	2655

Analysis and processing of national data

Estimation and forecasting

Categories	Forest Carbon (million tons)								
	1990	2000	2010	2015	2016	2017	2018	2019	2020
Carbon in above-ground biomass	3977	4732	5478	6261	6431	6567	6704	6841	6977
Carbon in below-ground biomass	1020	1223	1387	1576	1617	1649	1682	1714	1746
Carbon in dead wood	751	886	1058	1214	1248	1248	1248	1248	1248

Note: The carbon data in above- and below-ground biomass in 2016 was from the 9th NFI of China, and the others were calculated according to the mean carbon fraction of the 9th NFI, which is 0.4901 for above-ground biomass and 0.4832 for below-ground biomass. The carbon in dead wood were calculated using carbon fraction 0.47. The data of carbon in litter and soil are not available.

Reclassification into FRA 2020 categories

Reclassification is not necessary.

FRA categories	Forest carbon (tonnes/ha)								
	1990	2000	2010	2015	2016	2017	2018	2019	2020
Carbon in above-ground biomass	25.31	26.73	27.31	29.77	30.27	30.64	31.01	31.36	31.72
Carbon in below-ground biomass	6.49	6.91	6.91	7.49	7.61	7.70	7.78	7.86	7.94
Carbon in dead wood	4.78	5.01	5.28	5.77	5.87	5.82	5.77	5.72	5.67
Carbon in litter									
Soil carbon									

Soil depth (cm) used for soil carbon estimates	
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Comments

The data of carbon in litter and soil are not available.

3 Forest designation and management

3a Designated management objective

National Data

Data sources + type of data source eg NFI, etc

No	References to sources of information	Year(s)	Additional comments
1	State Forestry Administration, 1990, China Forest Resources Statistics (1984-1988)	1986	Data of 3 rd National Forest Inventory in China
2	State Forestry Administration, 1995, China Forest Resources Statistics (1989-1993)	1991	Data of 4 th National Forest Inventory in China
3	State Forestry Administration, 2000, China Forest Resources Statistics (1994-1998)	1996	Data of 5 th National Forest Inventory in China
4	State Forestry Administration, 2005, China Forest Resources Statistics (1999-2003)	2001	Data of 6 th National Forest Inventory in China
5	State Forestry Administration, 2010, China Forest Resources Statistics (2004-2008)	2006	Data of 7 th National Forest Inventory in China
6	State Forestry Administration, 2014, China Forest Resources Statistics (2009-2013)	2011	Data of 8 th National Forest Inventory in China
7	National Forestry and Grassland Administration, 2019, China Forest Resources Statistics (2014-2018)	2016	Data of 9 th National Forest Inventory in China

National classification and definitions

Categories	Definitions
Protection forest	Forest mainly for ecological protection, including water conservation forest, soil and water protection forest, windbreak and sand fixation forest, agricultural and pastoral shelter forest, bank protection forest, road protection forest, other shelterbelt forest.
Special-use forest	Forest mainly for tree species resources conservation, ecological environment protection, national defense, forest tourism and scientific experiments, including national defense forest, experimental forest, reserved-seed forest, environmental protection forest, scenic forest, historic site & revolutionary memorial forest, and nature conservation forest.
Timber forest	Forest mainly for timber or bamboo production.
Fuelwood forest	Forest mainly for fuelwood production.
Economic forest	Forest composed of economic species, of which main purpose is to produce fruits, cooking oil, drinks, spices, industrial raw materials and medicinal materials.

Note: According to Forest Law of China, all forests are classified into five categories above, no multiple use forest.

Original data

Categories		Forest area （1000 ha）						
		1986	1991	1996	2001	2006	2011	2016
Arbor forest, Bamboo forest, Open forest	Soil and water protection forest	17139.8	17390.2	16508.4	31694.5	45368.0	47061.8	46146.2
	Other protection forest	15618.1	15920.6	15485.2	29835.3	42754.5	44759.4	48933.4
	nature conservation forest	3172.8	3248.9	3109.6	4669.7	8250.0	10103.6	12442.9
	National defense forest	629.1	661.8	719.4	1128.7	2057.4	2455.4	2321.3

	Other special use forest	711.4	735.7	738.9	1121.5	1991.9	2161.3	2669.5
	Timber forest	88609.8	93182.0	104819.7	84179.7	65692.0	68356.8	73247.3
	Fuelwood forest	4603.4	4435.1	4510.2	3083.2	1786.5	1812.0	1256.0
	Economic forest	10094.1	11808.2	14784.5	15633.4	14914.3	15047.1	15307.2
Unestablished afforestation land		7288.1	7138.3	4615.1	4893.6	10461.8	6502.6	6991.4
Nursery land		184.5	114.9	122.5	270.9	454.0	506.4	719.8
Cut-over& fired-over forest land		4430.6	3669.6	3106.8	2610.1	2794.3	2865.4	2424.9

Analysis and processing of national data

Estimation and forecasting

Categories	Forest area （1000ha）						
	1990	1995	2000	2005	2010	2015	2020
Production	75481.7	84143.0	73903.3	55818.8	53651.2	56564.7	59246.9
Protection of soil & water	17340.1	16684.8	28665.4	42635.3	46723.0	46329.3	46457.5
Conservation of biodiversity	3233.7	3137.5	4357.7	7533.9	9732.9	11975.0	14120.1
Social services	730.8	738.3	1045.0	1817.8	2127.4	2567.9	2940.5
Multiple uses	45114.6	48265.5	55584.8	62185.2	65673.5	70762.4	75282.9
Other	655.3	707.9	1046.8	1871.7	2375.8	2348.1	2426.9
None/unknown	10990.1	8339.1	7547.3	12105.6	10145.6	9406.6	7880.4

Reclassification into FRA 2020 categories

1 Reclassification matrix

Classifications and definitions		FRA categories						
National classes		Production	Protection of soil and water	Conservation of biodiversity	Social services	Multiple uses	Other	None/ unknown
Arbor forest, Bamboos and Open forest	Protection of soil and water	0%	100%	0%	0%	0%	0%	0%
	Other protection	0%	0%	0%	0%	100%	0%	0%
	Nature conservation	0%	0%	100%	0%	0%	0%	0%
	National defense	0%	0%	0%	0%	0%	100%	0%
	Other special use	0%	0%	0%	100%	0%	0%	0%
	Timber	70%	0%	0%	0%	30%	0%	0%
	Fuelwood	100%	0%	0%	0%	0%	0%	0%
	Economic	80%	0%	0%	0%	20%	0%	0%

Unestablished afforestation land	0%	0%	0%	0%	0%	0%	100%
Nursery land	100%	0%	0%	0%	0%	0%	0%
Cut-over& fired-over forest land	0%	0%	0%	0%	0%	0%	100%

2 Reclassification result

FRA2020 categories	Forest area (1000ha)						
	1986	1991	1996	2001	2006	2011	2016
Production	74890.0	79224.0	89834.1	74786.6	60156.3	62205.8	65494.7
Protection of soil & water	17139.8	17390.2	16508.4	31694.5	45368.0	47061.8	46146.2
Conservation of biodiversity	3172.8	3248.9	3109.6	4669.7	8250.0	10103.6	12442.9
Social services	711.4	735.7	738.9	1121.5	1991.9	2161.3	2669.5
Multiple uses	44219.9	46236.8	49888.0	58215.9	65445.0	68275.9	73969.0
Other	629.1	661.8	719.4	1128.7	2057.4	2455.4	2321.3
None/unknown	11718.7	10807.9	7721.9	7503.7	13256.1	9368.0	9416.3

Primary designated management objective

FRA 2020 categories	Forest area (1000 ha)				
	1990	2000	2010	2015	2020
Production (a)	78 357.18	77 796.11	61 795.94	64 836.90	67 877.87
Protection of soil and water (b)	17 340.12	28 657.28	46 723.04	46 329.32	45 935.60
Conservation of biodiversity (c)	3 233.68	4 357.68	9 732.88	11 975.04	14 217.20
Social Services (d)	730.84	1 044.98	2 127.42	2 567.86	3 008.30
Multiple use (e)	45 833.44	56 550.31	67 709.68	72 830.40	77 951.11
Other (specify in comments) (f)	655.26	1 046.84	2 375.80	2 348.12	2 320.44
None/unknown (g)	10 990.07	7 547.35	10 145.62	9 406.61	8 667.66
Total forest area	157 140.59	177 000.55	200 610.38	210 294.25	219 978.18

Total area with designated management objective

FRA 2020 categories	Forest area (1000 ha)				
	1990	2000	2010	2015	2020
Production	108 330.52	107 381.14	85 147.20	89 568.70	93 990.20
Protection of soil and water	17 340.12	28 657.28	46 723.04	46 329.32	45 935.60
Conservation of biodiversity	3 233.68	4 357.68	9 732.88	11 975.04	14 217.20
Social Services	730.84	1 044.98	2 127.42	2 567.86	3 008.30
Other (specify in comments)	16 515.36	28 012.12	46 734.22	50 446.72	54 159.22

Comments

Other forest in the 1st table includes the national defense forest in the special-use forests, and other forest in the 2nd table includes the national defence forest in the special-use forests and other protection forests, including water conservation forest, windbreak and sand fixation forest, agricultural and pastoral shelter forest, sea- and river-bank protection forest, road protection forest, and other shelterbelt forest.

3b Forest area within protected areas and forest area with long-term management plans

National Data

Data sources + type of data source eg NFI, etc

No	References to sources of information	Year(s)	Additional comments
1	State Forestry Administration, 2014, FRA2015 Country Report -China.	1990-2010	Related assessment results
2	State Forestry Administration, 1991-2018, China Forestry Statistics Yearbook	1990-2017	Data of forest parks area

National classification and definitions

Categories	Definitions
Nature reserve	A certain area where determined by law to provide special protection and regulation for representative natural ecosystem, the natural concentrated area distribution with rare and endangered fauna and flora species, and the land, water inland or sea area where protect significant natural relic.
Forest park	A natural area with forest resources and forest landscape as the main body, approved in accordance with legal procedures, where the main function is to protect forest landscape resources, develop natural education and forest eco-tourism. It is a prohibited development area in the development of land space, and is one of the important protected areas in China.

Original data

1 Nature reserve

Category	Forest area (1000 ha)						
	1990	2000	2005	2010	2015	2016	2017
Forest area within nature reserve	4513	9100	12900	14900	15092	15092	15092

Note: The data in this table was provided by the Department of Natural Protected Areas Management of National Forestry and Grassland Administration (NFGA).The areas before 2015 were from the FRA2015 Country Report of China, and the areas in 2016 and 2017 were the same as that in 2015.

2 Forest park

Category	Forest area (1000 ha)					
	1990	2000	2010	2015	2016	2017
Forest area within forest park	125	7378	12583	13513	14150	15211

Note: The data in this table was provided by the Department of State-owned Forest Farms and Seedling Management of NFGA.

3 Forest area with long-term forest management plan

The overall trend of forest area with long-term forest management plan is upward. According to the results of thematic investigation and expert consultation, the forest areas with long-term forest management plan in 1990, 2000, 2010, 2015, and 2020 account for 50%, 55%, 60%, 65% and 70% of the total forest area, respectively.

Analysis and processing of national data

Estimation and forecasting

Categories	Forest area (1000 ha)									
	1990	2000	2010	2015	2016	2017	2018	2019	2020	

Forest area in nature reserve	4513	9100	14900	15092	15092	15092	15092	15092	15092
Forest area in forest park	125	7378	12583	13513	14150	15211	15228	15243	15258
Forest area within legally established protected areas	4638	16478	27483	28605	29242	30303	30320	30335	30350
Forest area with long-term forest management plan	78570	97351	120366	136691	140224	143607	147029	150488	153985

Note: The forest area in nature reserves after 2015 is the same as that in 2015, and the forest area in forest parks after 2017 is predicted to increase annually by about 15 thousands hectares. Forest areas in 1990, 2000, 2010, 2015, and 2020 with long-term forest management plan are estimated by 50%, 55%, 60%, 65% and 70% of total forest area for every year, respectively.

Reclassification into FRA 2020 categories

Forest area within legally established protected areas includes forest areas in nature reserve and forest park.

FRA categories	Area (1000 ha)								
	1990	2000	2010	2015	2016	2017	2018	2019	2020
Forest area within protected areas	4 638.00	16 478.00	27 483.00	28 605.00	29 242.00	30 303.00	30 320.00	30 335.00	30 350.00
Forest area with long-term forest management plan	78 570.00	97 351.00	120 366.00	136 691.00	140 224.00	143 607.00	147 029.00	150 488.00	153 985.00
...of which in protected areas									

Comments

4 Forest ownership and management rights

4a Forest ownership

National Data

Data sources + type of data source eg NFI, etc

No	References to sources of information	Year(s)	Additional comments
1	State Forestry Administration, 1990, China Forest Resources Statistics (1984-1988)	1986	Data of 3 rd National Forest Inventory in China
2	State Forestry Administration, 1995, China Forest Resources Statistics (1989-1993)	1991	Data of 4 th National Forest Inventory in China
3	State Forestry Administration, 2000, China Forest Resources Statistics (1994-1998)	1996	Data of 5 th National Forest Inventory in China
4	State Forestry Administration, 2005, China Forest Resources Statistics (1999-2003)	2001	Data of 6 th National Forest Inventory in China
5	State Forestry Administration, 2010, China Forest Resources Statistics (2004-2008)	2006	Data of 7 th National Forest Inventory in China
6	State Forestry Administration, 2014, China Forest Resources Statistics (2009-2013)	2011	Data of 8 th National Forest Inventory in China
7	National Forestry and Grassland Administration, 2019, China Forest Resources Statistics (2014-2018)	2016	Data of 9 th National Forest Inventory in China

National classification and definitions

Categories	Definitions
State-owned	Forest owned by the State.
Community-owned	Forest owned by the communities in countryside.
Private ownership	Forest owned by individuals, families, communities, private co-operatives, private corporations and other institutions.

Original data

National categories		Forest area （1000ha）						
		1986	1991	1996	2001	2006	2011	2016
Arbor forest, economic forest, bamboos, and open forest	State-owned	61053.3	64008.2	69781.5	74395.4	75456.9	76156.0	78399.1
	Community-owned	53054.3	55622.2	60639.1	64678.4	52780.3	35539.6	35357.2
	Private ownership	26470.9	27752.1	30255.3	32272.2	54577.4	80061.8	88567.5
Unestablished afforestation land, nursery land, and cut-over& fired-over forest land	State-owned	2552.0	2341.8	1681.8	1666.6	1547.9	1173.3	1131.3
	Community-owned	9351.2	8581.0	6162.6	6108.0	12162.2	8701.1	9004.8

Analysis and processing of national data

Estimation and forecasting

Categories	Forest area （1000 ha）					
	1990	1995	2000	2005	2010	2015

Private	27495.90	29754.70	31868.80	50116.40	74964.90	86866.40
Public	129644.70	136722.64	145131.72	142927.52	125645.46	123427.92

Reclassification into FRA 2020 categories

1 Reclassification matrix

Classifications and definitions		FRA categories		
National classes		Private	Public	Other
Arbor forest, bamboos, and open forest	State-owned	0%	100%	0%
	Community-owned	0%	100%	0%
	Private ownership	100%	0%	0%
Unestablished afforestation land, nursery land, and cut-over & fired-over forest land	State-owned	0%	100%	0%
	Community-owned	0%	100%	0%

2 Reclassification results

Categories	Forest area （1000 ha）						
	1986	1991	1996	2001	2006	2011	2016
Private	26470.90	27752.10	30255.30	32272.20	54577.40	80061.80	88567.50
Public	126010.80	130553.20	138265.00	146848.40	141947.30	121570.00	123892.40

FRA categories	Forest area (1000 ha)			
	1990	2000	2010	2015
Private ownership (a)	27 495.89	31 868.83	74 964.92	86 866.33
...of which owned by individuals				
...of which owned by private business entities and institutions				
...of which owned by local, tribal and indigenous communities				
Public ownership (b)	129 644.70	145 131.72	125 645.46	123 427.92
Unknown/other (specify in comments) (c)	0.00	0.00	0.00	0.00
Total forest area	157 140.59	177 000.55	200 610.38	210 294.25

Comments

4b Holder of management rights of public forests

National Data

Data sources + type of data source eg NFI, etc

Same as in 4a. In the NFI of China, two kinds of ownership were classified, that is, land ownership (State-owned/ Community-owned) and stock ownership (State-owned/Community-owned/ Private ownership). The forest ownership and management rights are consistent in data processing. The forest areas with growing stock, such as arbor forest, bamboo forest and open forest land, were classified according to stock ownership, and the forest areas without growing stock, such as unestablished afforestation land, nursery land, and cut-over & fired-over forest land, were classified according to land ownership. Therefore, when the public forest area needs to be classified according to management rights in table 4b, they would be all filled in public administration, no private ownership

National classification and definitions

Same as in 4a.

Original data

Same as in 4a.

Analysis and processing of national data

Estimation and forecasting

Same as in 4a.

Reclassification into FRA 2020 categories

Same as in 4a.

FRA categories	Forest area (1000 ha)			
	1990	2000	2010	2015
Public Administration (a)	129 644.70	145 131.72	125 645.46	123 427.92
Individuals (b)	0.00	0.00	0.00	0.00
Private business entities and institutions (c)	0.00	0.00	0.00	0.00
Local, tribal and indigenous communities (d)	0.00	0.00	0.00	0.00
Unknown/other (specify in comments) (e)	0.00	0.00	0.00	0.00
Total public ownership	129 644.70	145 131.72	125 645.46	123 427.92

Comments

5 Forest disturbances

5a Disturbances

National Data

Data sources + type of data source eg NFI, etc

No	References to sources of information	Year(s)	Additional comments
1	State Forestry Administration, 2001-2018, China Forestry Statistics Yearbook	2000-2017	The forest area reported to be disturbed by insect pests and diseases

National classification and definitions

Same as the FRA classification.

Original data

Categories	Forest area affected (1000 ha)								
	2000	2001	2002	2003	2004	2005	2006	2007	2008
Insects	ID	ID	6874.7	7184.6	7440.3	7492.9	8298.7	8877.2	8431.9
Diseases	ID	ID	810.1	757.5	757.9	1014.0	1038.7	1109.5	1168.3
Total	ID	ID	7684.8	7942.1	8198.2	8506.9	9337.3	9986.6	9600.2
Categories	2009	2010	2011	2012	2013	2014	2015	2016	2017
Insects	8503.0	8523.2	8459.1	8462.9	8474.6	8412.8	8466.4	8570.2	9059.7
Diseases	1031.2	1290.6	1197.2	1311.6	1391.7	1372.8	1390.5	1388.9	1330.9
Total	9534.2	9813.8	9656.3	9774.5	9866.3	9785.6	9856.9	9959.1	10390.5

Note: The data in this table was provided by Department of Ecological Conservation and Restoration of NFGA. There are no separate data of forest area affected by insects and diseases in 2000 and 2001 , only the total area affected by insects, diseases and rats. In addition, there are no data of forest area affected by severe weather events.

Analysis and processing of national data

Estimation and forecasting

No need.

Reclassification into FRA 2020 categories

Not necessary.

FRA categories	Area (1000 ha)																	
	2000	2001	2002	2003	2004	2005	2006	2007	2008	2009	2010	2011	2012	2013	2014	2015	2016	2017
Insects (a)			6 875.00	7 185.00	7 440.00	7 493.00	8 298.00	8 877.00	8 432.00	8 503.00	8 523.00	8 459.00	8 463.00	8 474.00	8 413.00	8 466.00	8 570.00	9 060.00
Diseases (b)			810.00	757.00	758.00	1 014.00	1 039.00	1 110.00	1 168.00	1 031.00	1 291.00	1 197.00	1 312.00	1 392.00	1 373.00	1 391.00	1 389.00	1 331.00
Severe weather events (c)																		
Other (specify in comments) (d)																		
Total (a+b+c+d)	–	–	7 685.00	7 942.00	8 198.00	8 507.00	9 337.00	9 987.00	9 600.00	9 534.00	9 814.00	9 656.00	9 775.00	9 866.00	9 786.00	9 857.00	9 959.00	10 391.00
Total forest area	177 000.55	179 120.60	–	–	–	–	196 524.70	–	–	–	200 610.38	201 631.80	–	–	–	210 294.25	212 459.87	214 339.47

Comments

There are no separate data of forest area affected by insects and diseases in 2000 and 2001, only the total area affected by insects, diseases and rats. In addition, there are no data of forest area affected by severe weather events.

5b Area affected by fire

National Data

Data sources + type of data source eg NFI, etc

No	References to sources of information	Year(s)	Additional comments
1	State Forestry Administration, 2000-2017, Forest Fire Statistics Report	2000-2017	Burned area and affected forest area

National classification and definitions

Same as the FRA classification.

Original data

Categories	Forest area affected (1000 ha)								
	2000	2001	2002	2003	2004	2005	2006	2007	2008
Burned area	169.45	192.53	131.82	1123.75	344.21	290.63	562.30	125.13	184.50
Forest area	90.12	46.18	47.63	451.02	142.24	73.70	408.26	29.29	52.54
Categories	2009	2010	2011	2012	2013	2014	2015	2016	2017
Burned area	213.64	116.24	63.42	43.17	42.89	55.34	33.08	18.16	44.43
Forest area	46.16	45.80	26.95	13.95	13.72	19.11	12.94	6.22	24.50

Note: The data in this table was provided by Department of Forest Public Security of NFGA.

Analysis and processing of national data

Estimation and forecasting

No need.

Reclassification into FRA 2020 categories

Not necessary.

FRA categories	Area (1000 ha)																	
	2000	2001	2002	2003	2004	2005	2006	2007	2008	2009	2010	2011	2012	2013	2014	2015	2016	2017
Total land area affected by fire	169.45	192.53	131.82	1 123.75	344.21	290.63	562.30	125.13	184.50	213.64	116.24	63.42	43.17	42.89	55.34	33.08	18.16	44.43
...of which on forest	90.12	46.18	47.63	451.02	142.24	73.70	408.26	29.29	52.54	46.16	45.80	26.95	13.95	13.72	19.11	12.94	6.22	24.50

Comments

5c Degraded forest

Does your country monitor area of degraded forest		No
If "yes"	What is the national definition of "Degraded forest"?	
	Describe the monitoring process and results	

Comments

6 Forest policy and legislation

6a Policies, Legislation and national platform for stakeholder participation in forest policy

National Data

Data sources + type of data source eg NFI, etc

See the comments below.

National classification and definitions

Same as the FRA classification.

Original data

No data.

Indicate the existence of	Boolean (Yes/No)	
	National	Sub-national
Policies supporting SFM	Yes	Yes
Legislations and regulations supporting SFM	Yes	Yes
Platform that promotes or allows for stakeholder participation in forest policy development	Yes	Yes
Traceability system(s) for wood products	No	No

Comments

6a.1 Policies that support sustainable forest management

Chinese government attaches great importance to forest management, and has formulated and issued a series of major decisions to support forestry construction and promote sustainable forest management. In 1981, the CPC Central Committee and the State Council issued the *Decision on Several Issues Concerning Protection of Forest and Forestry Development*. Since 1987, the system of forest cutting quota management has been established to implement quota management for wood harvest, aiming at controlling the total forest resources consumption and its structure.

Since the end of the last century, key forestry projects, such as protection of natural forests, construction of shelter forests, returning farmland to forests, Beijing-Tianjin sandstorm source control, conservation of wildlife, construction of nature reserves, establishment of fast-growing and high-yielding forestry bases, have been initiated and implemented. The compensation system of forest eco-function has been established. Specific administration systems, management mechanisms and policies have been respectively applied to public welfare forests and commercial forests. In 2003, the CPC Central Committee and the State Council promulgated the *Decision on Accelerating the Development of Forestry* and established a strategy for sustainable development of forestry with emphasis on ecological construction. In 2008, the *Opinions on Comprehensively Promoting the Collective Forest Tenure Reform* was promulgated, which significantly liberated and improved the productivity of collective forest land in rural areas. In 2010, the *Outline of Forestland Protection and Utilization Plan (2010-2020)* was enacted, forestland protection and management has been comprehensively strengthened by establishment of utilization control system and logging quota management. The red line of forest and forestland was delimited in 2013, and responsibility of forestry ecological protection red line is created to develop forestry and improve ecological environment. In 2016, the *National Forest Resources Management Plan (2016-2050)* was drafted, which put forward the guiding ideology, basic principles, objectives and tasks, management layout, management strategy, technical system and construction scale of national forest management in the next 35 years, and safeguarding policies and measures were proposed.

Furthermore, relevant policies and regulations such as *Opinion on Reform and Improvement of Collective Forest Harvesting Management*, *Measure on National Public Welfare Forest Zoning and Delimitation*, *Measure on National Public Welfare Forest Management*, *Opinion on Further Revitalizing Collective Forest Management Rights* and other sustainable forest management policies have been promulgated. Local governments have also issued corresponding policies and measures to accelerate sustainable forest management at local levels. The promulgation and implementation of a series of major policies and measures has greatly driven forest protection, management and administration in China, to improve forest productivity. Achievement of growth of forest resources, both in areas and in stock volumes, has been continuously made, the income of forest farmer has been enhanced, better ecology and social harmony has been achieved.

6a.2 Legislation and/or regulations about sustainable forest management

Since the reform and opening up, along with the establishment and improvement of the socialist market economic system and national legislation construction, forestry legislation has gradually established. In 1979, the *Forest Law (Trail)* was reviewed and adopted at the 6th meeting of the 5th National People's Congress Standing Committee, which is a symbol of the conversion of forest management from the simply relying on the administrative approaches to use of legal and administrative approaches for adjustment, regulation, promotion and safeguard of forestry healthy development. In 1981, the 4th Meeting of the 5th National People's Congress reviewed and adopted the *Resolution on the Unfolding of a Nationwide Voluntary Tree-planting Campaign*. The Forest Law was discussed and passed at the 6th Meeting of the National People's Congress Standing Committee in 1984, and amended and adopted by the 2nd Meeting of the 9th National People's Congress Standing Committee in 1998. In 2000, the *Regulations for the Implementation of the Forest Law* was enacted. According to changes in China's economy and society, in order to adapt the needs of sustainable forest development, China is considering to further amend the *Forest Law*.

In addition, China has also successively enacted a series of laws and regulations which are closely relevant to forestry, including *Land Management Law*, *Seed Law*, *Combating Desertification Law*, *Law on Contracting of Rural Land*, *Agriculture Law*, *Law of Promoting Agricultural Technology*, *Law on Specialized Farmers Cooperative* as well as regulations on *Nature Reserve*, *Import/Export Management of Endangered Wildlife*, *Wild Plants Protection*, *Plant Quarantine*, *Forest Fire Prevention* and etc.

According to China's legislative system, all provinces, autonomous regions and municipalities can formulate local regulations and rules without contradicting laws and administrative regulations. In light of the actual needs, local governments have formulated a large number of local regulations and rules, which have enhanced the operability of laws and administrative regulations.

In general, China has established a relatively complete legal system in terms of tree seedling management, forestland and forest protection, forest harvesting, public welfare forest and natural forest protection, wildlife protection, plant quarantine, forest pest control and forest fire prevention. It provides a sound legal safeguard for the achievement of forest sustainable management and promotion of ecological civilization.

6a.3 Platform that promotes or allows stakeholder participation in forest policy development

The Chinese government has established a platform for the participation among relevant ministries of the State Council, a platform for the participation of local forestry authorities at all levels, and a platform for public participation, etc. Comments and appeals from stakeholders have been extensively collected in the process of development and enactment of national laws and regulations, departmental rules, and major forestry policies. In particular, for extremely important issues, various workshops and communication meetings are also held to fully listen to the appeals of various stakeholders.

Furthermore, the National Forestry and Grassland Administration (NFGA) has set up an information disclosure column on its official website (<http://www.forestry.gov.cn>), to disclose information on government affairs, publicize policies on forestry reform and development, also accept public advice, and provide online consulting services. With regard to the information outside of NFGA voluntarily disclosed, the public can fill out the information request form and submit an application in written form. The responsible agencies of NFGA will reply with legal provisions to respond the appeals of stakeholders and accept the public supervision.

6a.4 Wood products traceability system

At present, China has not yet established a wood products traceability system.

6b Area of permanent forest estate

National Data

Data sources + type of data source eg NFI, etc

None.

National classification and definitions

None.

Original data

No data.

FRA 2020 categories	Forest area (1000 ha)					
	Applicable?	1990	2000	2010	2015	2020
Area of permanent forest estate	No					

Comments

The permanent forest estate concept does not exist at present in China.

7 Employment, education and NWFP

7a Employment in forestry and logging

National Data

Data sources + type of data source eg NFI, etc

No	References to sources of information	Year(s)	Additional comments
1	State Forestry Administration, 1990-2017, China Forestry Statistics Yearbook	1989-2016	Data of employment in forestry

National classification and definitions

Categories	Definitions
Employment in forestry	The term refers to the employees of all forestry units managed by forestry departments, including those employed by forestry enterprises, institutions and government units.
...of which silviculture and other forestry activities	This class includes employment in tree breeding and seedling, silviculture and other forestry activities.
...of which logging	This class includes employment in logging and transportation of timbers and bamboos.
...of which gathering of non wood forest products	This class includes employment in planting and collection of economic forest products, mostly for business.
...of which support services to forestry	This class includes employment in wildlife protection, forestry engineering technology and planning management, forestry science and technology exchange and promotion, forestry public management, forest fire prevention and pest control and other forestry production services.

Original data

National categories	1989		1990		1991		1999		2000		2001	
	Total	Female	Total	Female	Total	Female	Total	Female	Total	Female	Total	Female
Employment in forestry	ID	ID	2489414	ID	2544892	ID	2212298	706731	1978195	622527	1823332	570019
...of which silviculture and other forestry activities	ID	ID	694111	ID	720417	ID	649304	207751	600963	193721	591704	189862
...of which logging	ID	ID	1071569	ID	1081834	ID	812035	248310	683048	207719	614007	187239
...of which gathering of non wood forest products	ID	ID	ID	ID	ID	ID	ID	ID	ID	ID	ID	ID
...of which support services to forestry	ID	ID	224103	ID	241470	ID	355946	89559	378832	96374	354295	87293
National categories	2009		2000		2001		2014		2015		2016	
	Total	Female	Total	Female	Total	Female	Total	Female	Total	Female	Total	Female
Employment in forestry	1315642	360013	1347142	357625	1312749	358348	1103914	308729	1157572	319517	1180371	335000
...of which silviculture and other forestry activities	482078	136695	487383	137163	503533	145321	570852	161315	556448	161708	562773	171589
...of which logging	418851	112294	444281	110790	427271	113709	185739	50271	256374	64811	279523	70120
...of which gathering of non wood forest products	ID	ID	ID	ID	ID	ID	4731	1675	5239	1875	5698	2221
...of which support services to forestry	342109	89002	353062	89921	330394	84355	290695	75572	294250	74931	278047	70761

Note: The data in this table was provided by Department of Planning and Finance of NFGA. There are no data in 1989, thus the reporting data in 1990 are the averages of two years, 1990 and 1991. There are no data of employment in gathering of non wood forest products until the recent years.

FRA 2020 categories	Full-time equivalents (1000 FTE)											
	1990			2000			2010			2015		
	Total	Female	Male	Total	Female	Male	Total	Female	Male	Total	Female	Male
Employment in forestry and logging	2 517.20			2 004.60	633.10	1 371.50	1 325.20	358.70	966.50	1 147.30	321.10	826.20
...of which silviculture and other forestry activities	707.30			614.00	197.10	416.90	491.00	139.70	351.30	563.40	164.90	398.50
...of which logging	1 076.70			703.00	214.40	488.60	430.10	112.30	317.90	240.50	61.70	178.80
...of which gathering of non wood forest products												
...of which support services to forestry	232.80			363.00	91.10	271.90	341.90	87.80	254.10	287.70	73.80	213.90

Comments

There no data in 1989, thus the reporting data in 1990 are the averages of two years, 1990 and 1991. There no data of employment in gathering of non wood forest products until the recent years.

7b Graduation of students in forest-related education

National Data

Data sources + type of data source eg NFI, etc

No	References to sources of information	Year(s)	Additional comments
1	State Forestry Administration, 1990-2017, China Forestry Statistics Yearbook	1989-2016	Data of education in forestry

National classification and definitions

Same as the FRA classification.

Original data

FRA2020 categories	Number of graduated											
	1989	1990	1991	1999	2000	2001	2009	2010	2011	2014	2015	2016
Doctoral degree	7	11	23	94	97	111	568	598	666	1055	637	580
Master's degree	331	280	292	283	371	420	4166	4800	4888	6324	6643	6505
Bachelor's degree	4700	4578	4493	5030	5826	7218	32405	32698	35182	40739	47384	50087
Technician certificate /diploma	7145	7483	11650	24481	27506	30141	48226	63809	71991	105414	102222	91405

Note: The data in this table was provided by Department of Human Resources of NFGA.The breakdown of number of graduated by gender is not available.

FRA 2020 categories	Number of graduated students											
	1990			2000			2010			2015		
	Total	Female	Male	Total	Female	Male	Total	Female	Male	Total	Female	Male
Doctoral degree	14.00			101.00			611.00			757.00		
Master's degree	301.00			158.00			4 618.00			6 491.00		
Bachelor's degree	4 590.00			6 025.00			33 428.00			46 070.00		
Technician certificate / diploma	8 759.00			27 376.00			61 342.00			99 680.00		
Total												

Comments

The breakdown of number of graduated by gender is not available.

7c Non wood forest products removals and value 2015

National Data

Data sources + type of data source eg NFI, etc

No	References to sources of information	Year(s)	Additional comments
1	State Forestry Administration, 2016, China Forestry Statistics Yearbook	2015	Quantity and value of non wood forest products

National classification and definitions

Same as the FRA classification.

Original data

Rank	Name of NWFP	Key species	Quantity	Unit	Value (1000 currency)	NWFP categories
1 st	Walnut	ID	3331703	Tone	ID	No.1
2 nd	Camellia seed	ID	2163492	Tone	ID	No.1
3 rd	Bamboos	ID	2354660	1000Stems	ID	No.5
4 th	Chinese chestnut	ID	2342054	Tone	ID	No.1
5 th	Rosin	ID	1326292	Tone	ID	No.7
6 th	Matrimony vine	ID	293220	Tone	ID	No.1
7 th	Bancoul nut	ID	412042	Tone	ID	No.8
8 th	Chinese red pepper	ID	398224	Tone	ID	No.1
9 th	Raw lacquer	ID	22806	Tone	ID	No.7
10 th	Gallnut	ID	25081	Tone	ID	No.3
All other plant products		ID	ID	ID	ID	
All other animal products		ID	ID	ID	ID	
Total		ID	11441338	Tone	ID	

Note: The data in this table was provided by Department of Planning and Finance of NFGA. The data of all values, animal products, and other plant products are not available.

	Name of NWFP product	Key species	Quantity	Unit	Value (1000 local currency)	NWFP category
#1	Walnut		3 331 703	Tone		1 Food
#2	Camellia seed		2 163 492	Tone		1 Food
#3	Bamboos		2 354 660	1000Stems		5 Raw material for utensils handicrafts construction
#4	Chinese chestnut		2 342 054	Tone		1 Food
#5	Rosin		1 326 292	Tone		7 Exudates
#6	Matrimony vine		293 220	Tone		1 Food
#7	Bancoul nut		412 042	Tone		8 Other plant products
#8	Chinese red pepper		398 224	Tone		1 Food
#9	Raw lacquer		22 806	Tone		7 Exudates
#10	Gallnut		25 081	Tone		3 Raw material for medicine and aromatic products
All other plant products						
All other animal products						
Total					—	

Name of currency	
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Comments

8 Sustainable Development Goal 15

8a Sustainable Development Goal 15

SDG Indicator 15.1.1 Forest area as proportion of total land area 2015

Indicator	Percent							
	2000	2010	2015	2016	2017	2018	2019	2020
Forest area as proportion of total land area 2015	18.78	21.29	22.31	22.54	22.74	22.94	23.14	23.34

Name of agency responsible	Department of Forest Resources Management, National Forestry and Grassland Administration
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SDG Indicator 15.2.1 Progress towards sustainable forest management

Sub-Indicator 1	Percent						
	2000-2010	2010-2015	2015-2016	2016-2017	2017-2018	2018-2019	2019-2020
Forest area annual net change rate	1.26	0.95	1.02	0.88	0.87	0.86	0.85

Name of agency responsible	Department of Forest Resources Management, National Forestry and Grassland Administration
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Sub-Indicator 2	Forest biomass (tonnes/ha)							
	2000	2010	2015	2016	2017	2018	2019	2020
Above-ground biomass stock in forest	54.54	55.72	60.75	61.76	62.52	63.26	64.00	64.72

Name of agency responsible	Department of Forest Resources Management, National Forestry and Grassland Administration
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Sub-Indicator 3	Percent (2015 forest area baseline)							
	2000	2010	2015	2016	2017	2018	2019	2020
Proportion of forest area located within legally established protected areas	7.84	13.07	13.60	13.91	14.41	14.42	14.43	14.43

Name of agency responsible	Department of Forest Resources Management, National Forestry and Grassland Administration
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Sub-Indicator 4	Percent (2015 forest area baseline)							
	2000	2010	2015	2016	2017	2018	2019	2020
Proportion of forest area under long-term forest management plan	46.29	57.24	65.00	66.68	68.29	69.92	71.56	73.22

Name of agency responsible	Department of Forest Resources Management,Department of Natural Protected Areas Management,National Forestry and Grassland Administration
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Sub-Indicator 5	Forest area (1000 ha)							
	2000	2010	2015	2016	2017	2018	2019	2020
Forest area under independently verified forest management certification schemes	0.00	1 377.75	6 774.11	6 527.08	6 755.36	7 589.20	—	—